

Advanced DBMS

Course code: M23DES211

**School of Computer Science and
Applications**

Abhay Kumar Srivastav

LECTURE -1

Agenda

- Importance of the subject
- Prerequisites
- Objectives
- Course Content
- Course Outcome
- Data
- Database
- DBMS
- RDBMS
- DBMS - Storage System



IMPORTANCE OF THE COURSE

1. The DBMS implies integrity constraints to get a high level of protection against prohibited access to data
2. DBMS offers a variety of techniques to store & retrieve data
3. DBMS serves as an efficient handler to balance the needs of multiple applications using the same data
4. A DBMS uses various powerful functions to store and retrieve data efficiently.



PREREQUISITES

1. Basic Knowledge of Database
2. database relationships
3. memory management in OS



OBJECTIVES

- 1 To understand hard disk structure, storage concept and File Organisation methods.**
- 2 To understand SQL Query translation mechanism, Algorithms selection process ,Cost Estimation of Algorithms and Optimization process.**
- 3 To understand Transaction processing concept, states, properties, and conflict control mechanism.**
- 4 To understand information retrieval mechanism and Web search engine as application.**



COURSE CONTENT

UNIT 1:File Storage and File Organization (10 Hrs.)

Storage and File Structure Overview of Physical Storage Media , Magnetic Disk(T2), Buffering of Blocks, Placing of File records on disk (T1), RAID (T2) .

File Organization , Indexes and HashingComparison of Three File Organizations -Heap Files, Sorted Files, Hashed Files , Choosing a File Organization; Overview of Indexes - Alternatives for Data Entries in an Index, Properties of Indexes , ISAM – ex: static index structure, B+ Tree – ex: dynamic index structure; Hashing – Introduction to Static Hashing and Dynamic Hashing (T3).

UNIT 2:Algorithms for Query Processing and Optimization

Introduction to Translating SQL Queries into Relational Algebra , Algorithms for External Sorting ; Algorithms for SELECT and JOIN Operations , Combining Operations Using Pipelining, Using Heuristics in Query Optimization ; Using Selectivity and Cost Estimates in Query Optimization (T1).



COURSE CONTENT CONTD..

UNIT 3: Transaction Processing System & Schedules

1. Introduction to Transaction Concept , Transaction Processing and System: Transaction Concept, A Simple Transaction Model, Storage Structure (T2); Introduction to Transaction Processing : Single-User versus Multiuser Systems, Why Concurrency Control Is Needed, Why Recovery Is Needed ; Transaction and System Concepts : Transaction States and Additional Operations; Desirable Properties of Transactions. **Concurrency Control through Schedules:** Taxonomy of Schedules – Characterizing Schedules Based on Recoverability, Characterizing Schedules Based on Serializability (T1); Transaction Isolation Levels (T2).

UNIT 4: System Architecture ,Information Retrieval and Web Search

System Architecture: Distributed databases & Parallel databases Database System Architecture (T2); Parallel versus Distributed database Architecture; Types of Distributed databases Systems - Homogenous, Heterogenous; Distributed database Storage-Data Fragmentation, Replication and Allocation techniques for Distributed database design ; Query Processing and Optimization in Distributed databases(T1)**Introduction to Information Retrieval and Web Search :** Information Retrieval (IR) Concepts : Introduction to Information Retrieval, Databases and IR Systems Comparison, Modes of IR System – Retrieval mode & Browsing mode, Generic IR Pipeline Framework ; Taxonomy of Retrieval Models -types ; Web Search and Analysis(T1).

COURSE OUTCOME

- 1 To understand hard disk structure, storage concept and File Organisation methods.**
- 2 To understand SQL Query translation mechanism, Algorithms selection process ,Cost Estimation of Algorithms and Optimization process.**
- 3 To understand Transaction processing concept, states, properties, and conflict control mechanism.**
- 4 To understand information retrieval mechanism and Web search engine as application.**



TEXT BOOKS AND REFERENCE BOOKS

Text book/s:

1. Ramez Elmasri, Sham Navathe .Fundamentals of Database Systems, Publisher: Pearson/Addison , 7Th Edition , ISBN-10: 0-13-397077-9 ISBN-13: 978-0-13-397077-7
2. Abraham Silberschatz, Henry F. Korth, S. Sudarshan ,Database System Concepts, Publisher: McGraw-Hill, 2011.
3. Raghu Ramakrishnan, Johannes Gehrke ,Database Management Systems - Publisher: McGraw-Hill ,2000.
- 4.

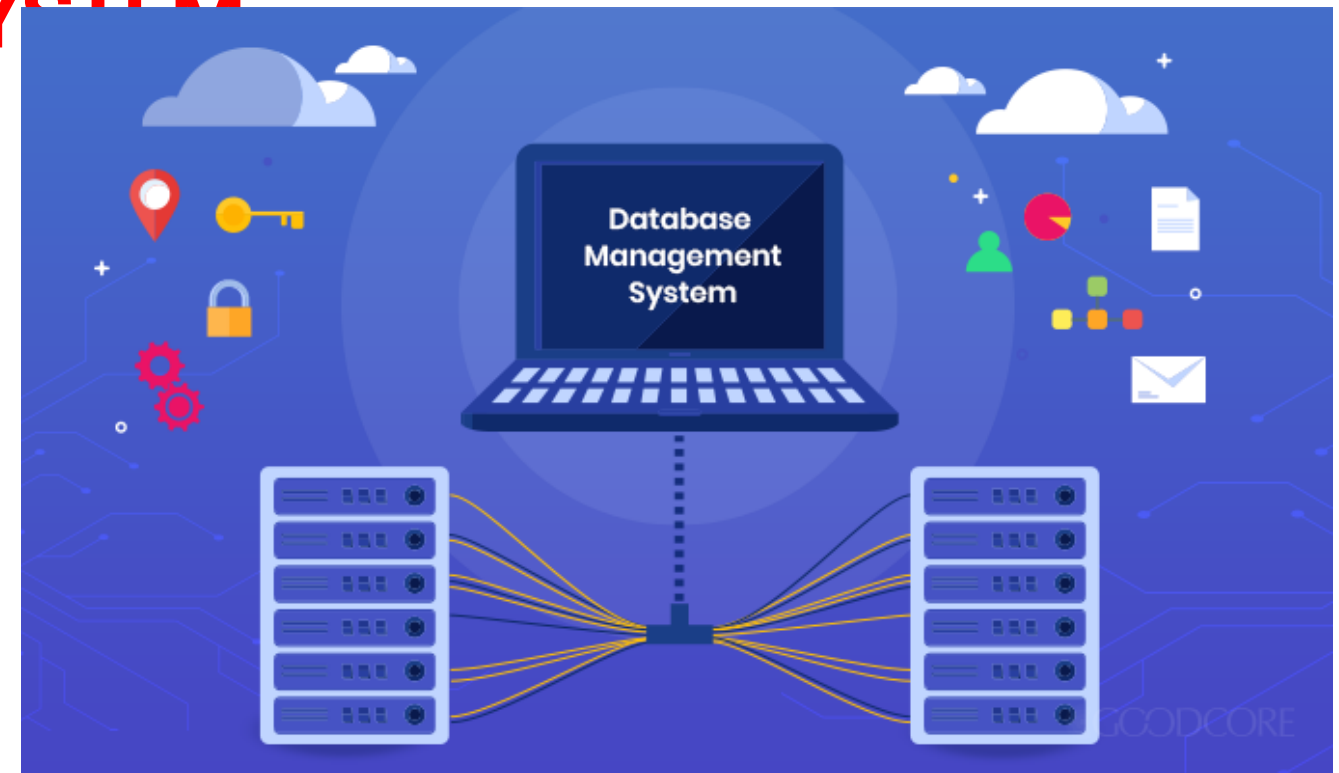
References:

- 1 Foster, Elvis C., Godbole, Shripad. Database Systems: A Pragmatic Approach. United States: Apress, 2014.
- 2 Rauber, Thomas., Rünger, Gudula. Parallel Programming: For Multicore and Cluster Systems. Germany: Springer Berlin Heidelberg, 2013.
- 3 Bochmann, G. von. Concepts for Distributed Systems Design. Germany: Springer Berlin Heidelberg, 2012.



DEFINITION

- Data
- Database
- A DATABASE MANAGEMENT SYSTEM
- Drawback of File System

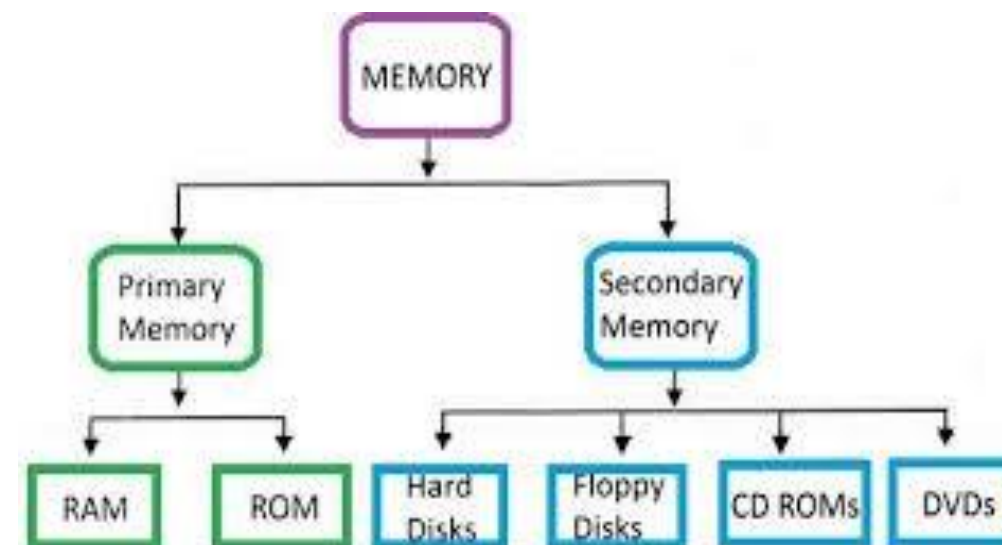


DBMS - STORAGE SYSTEM

Databases are stored in file formats, which contain records. At physical level, the actual data is stored in electromagnetic format on some device. These storage devices can be broadly categorized into three types –

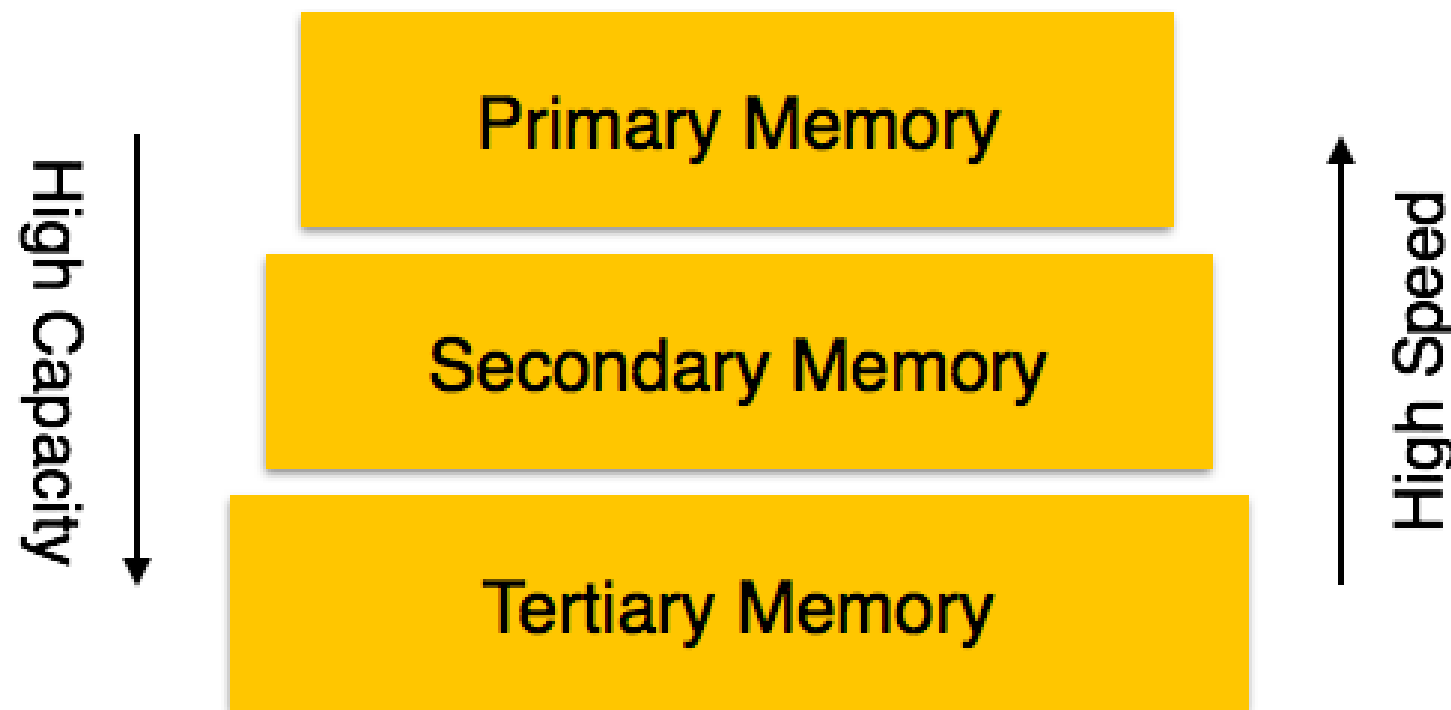
Primary Storage - The memory storage that is directly accessible to the CPU comes under this category.

Secondary Storage – Secondary storage devices are used to store data for future use or as backup.



DBMS - STORAGE SYSTEM

Tertiary Storage – Tertiary storage is used to store huge volumes of data.



MAGNETIC DISK STRUCTURE

Most of the secondary storage is in the form of magnetic disks.

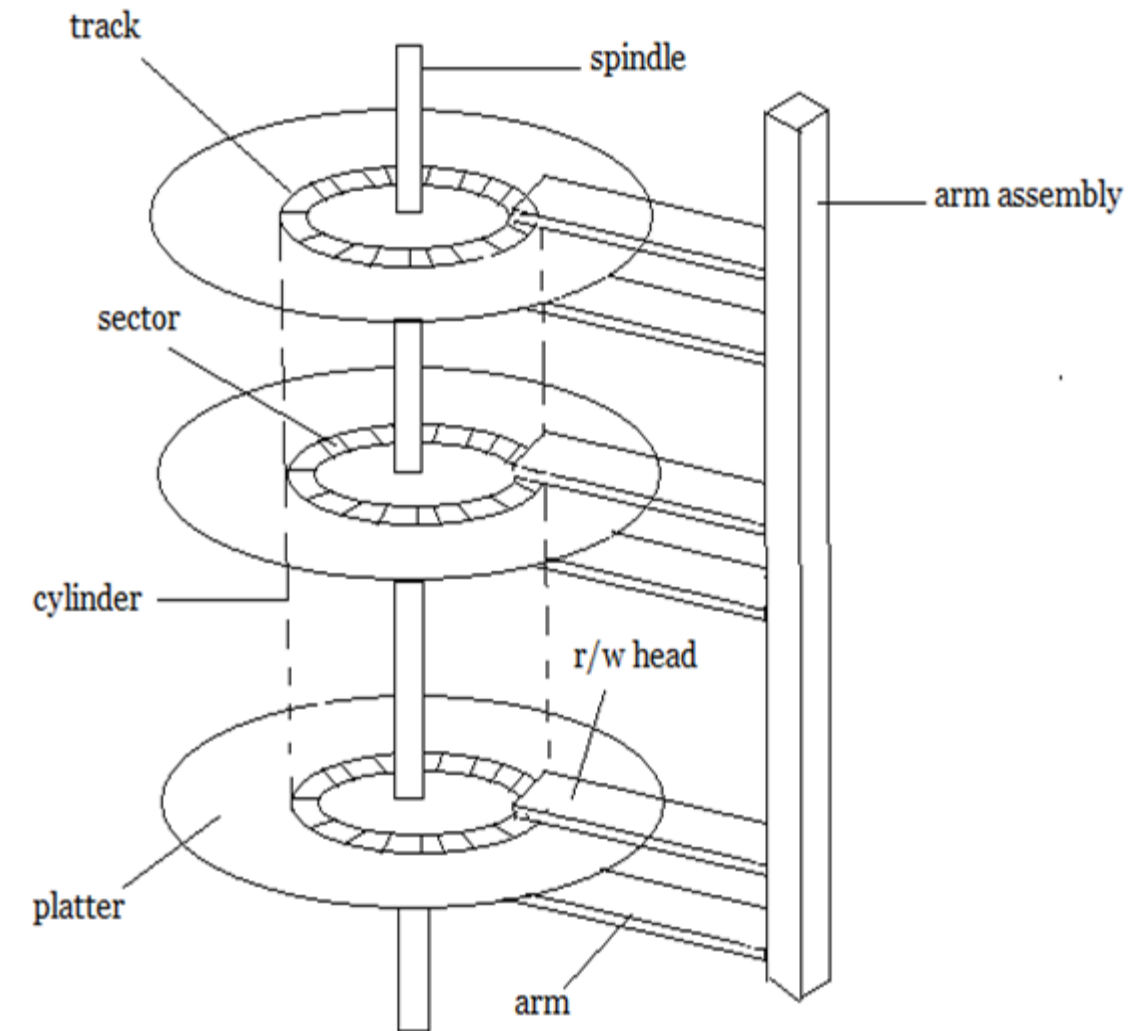
- A magnetic disk contains several **platters**.
- Each platter is divided into circular shaped **tracks**.
- The length of the tracks near the centre is less than the length of the tracks farther from the center. Each track is further divided into **sectors**



MAGNETIC DISK STRUCTURE

The speed of the disk is measured as two parts:

- **Transfer rate:** This is the rate at which the data moves from disk to the computer.
- **Random access time:** It is the sum of the seek time and rotational latency.
- **Seek time** is the time taken by the arm to move to the required track



Structure of a magnetic disk



SUMMARY

- Data
- Database
- DBMS
- RDBMS
- DBMS - Storage System



QUIZ

1-The storage device that uses rigid, permanently installed magnetic disks to store data is

1. Floppy
2. Permanent disk
3. Optical disk
4. Hard disk

ANSWER: Hard disk

2-Which of the following is the secondary storage device that uses a long plastic strip coated with a magnetic material as recording medium?

1. Compact disk
2. Hard disk
3. Magnetic tape
4. None of the above

ANSWER: Magnetic tape



LECTURE -2



- **DATABASE**
- **DATABASE MANAGEMENT SYSTEM**
- **Storage System**

OBJECTIVE

- **Disk Array**
- **Data striping**
- **Reliability**
- **RAID**
- **Redundancy**



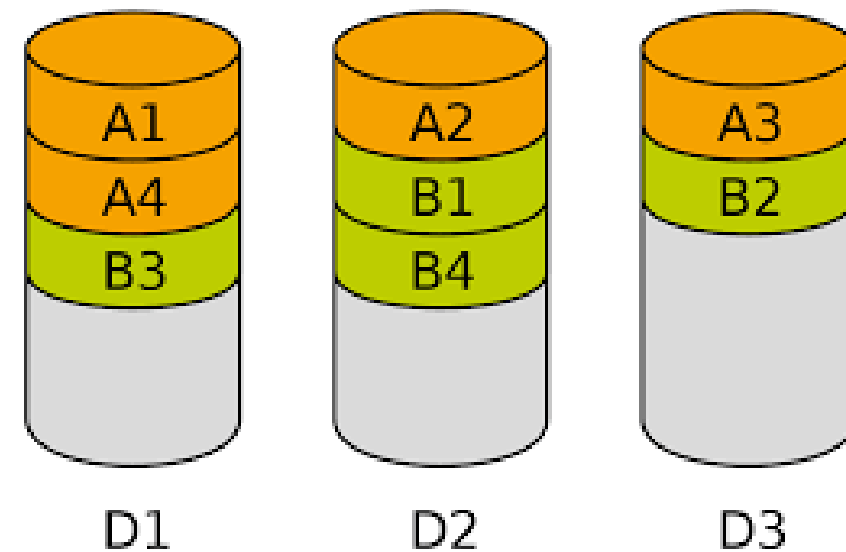
DISK ARRAY

- Is an arrangement of several disks, organized to increase performance and improve reliability of the resulting storage system.
- Performance is increased through data.



DATA STRIPING

- distributes data over several disks to give the impression of having a single large, very fast disk.



MEMORY HIERARCHY: RAID (REDUNDANT ARRAYS OF INDEPENDENT DISKS)

- **Reliability** is improved through **redundancy**. Instead of having a single copy of the data, redundant information is maintained.
- Disk arrays that implement a combination of data striping and redundancy are called redundant arrays of independent disks, or in short, **RAID**.



REDUNDANCY

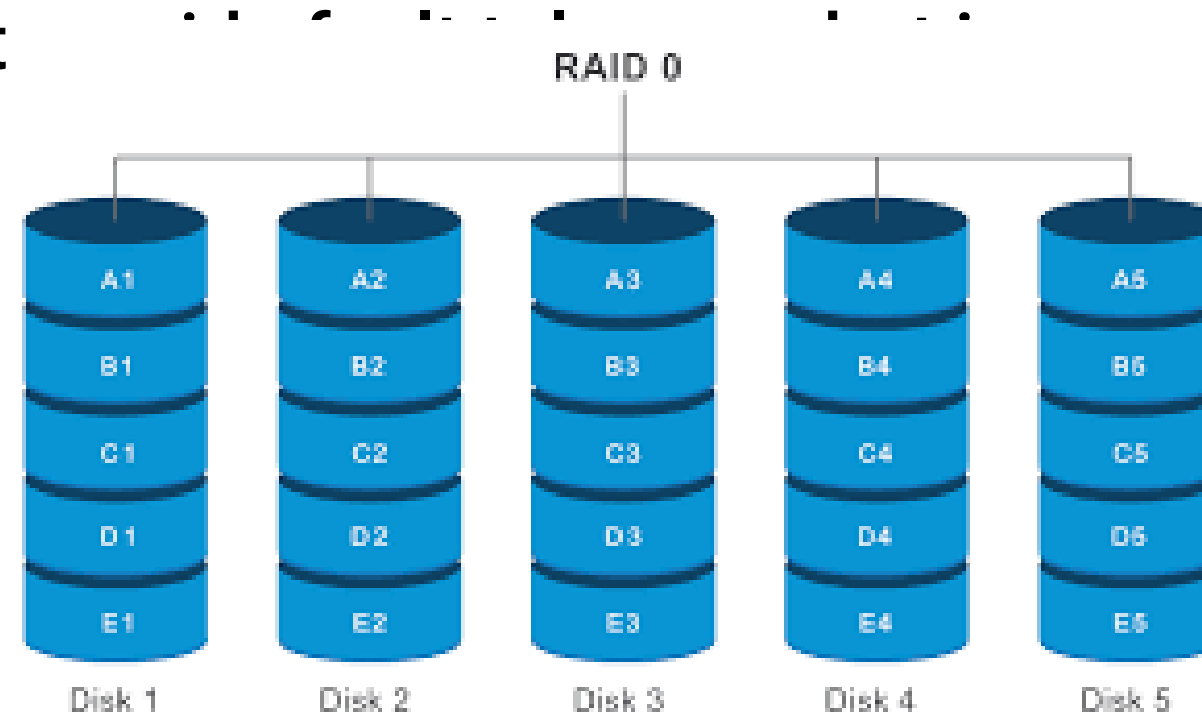
- Reliability of a disk array can be increased by storing redundant information. If a disk fails, the redundant information is used to reconstruct the data on the failed disk. Redundancy can immensely increase the MTTF(Mean-time-to-failure) of a disk array.
- When incorporating redundancy into a disk array design, we have to make two choices.
- We can either store the redundant information on a small number of check disks or distribute the redundant information uniformly over all disks.



LEVELS OF REDUNDANCY

Level 0: Nonredundant

- RAID level 0 provides data stripping, i.e., a data can place across multiple disks. It is based on stripping that means if one disk fails then all data in the array is lost.
- This level doesn't **as the system** performance.



LEVELS OF REDUNDANCY

Advantages of RAID 0:

- Throughput is increased because multiple data requests probably not on the same disk.
- Full utilizes the disk space and provides high performance.
- It requires minimum 2 drives.

Disadvantages of RAID 0:

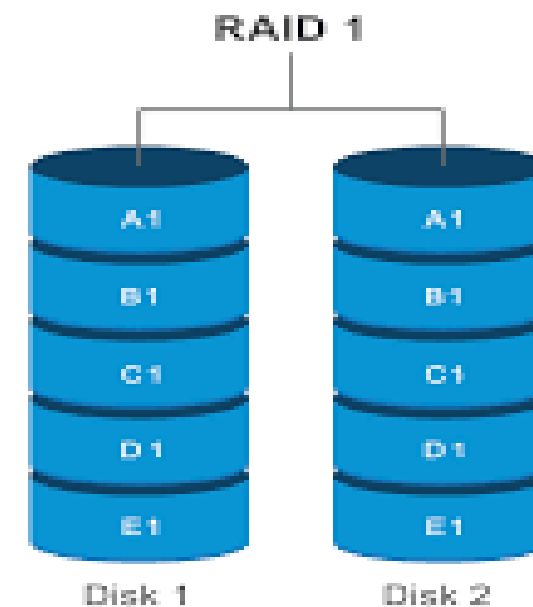
- It doesn't contain any error detection mechanism.
- The RAID 0 is not a true RAID because it is not fault-tolerance.
- Failure of either disk results in complete data loss in respective array.



LEVELS OF REDUNDANCY:

Level 1: Mirrored

- This level is called mirroring of data as it copies the data from drive 1 to drive 2. It provides 100% redundancy in case of a failure.
- Only half space of the drive is used to store the data. The other half of drive is just a mirror to the already stored data.



LEVELS OF REDUNDANCY:

Level 1: Mirrored

Advantages of RAID 1:

- Fault tolerance
 - if one disk fails, then the other automatically takes over.
 - the array will function even if any one of the drives fails.

Disadvantages of RAID 1:

- One extra drive is required per drive for mirroring, so the expense is higher.



SUMMARY

- Disk Array
- Data striping
- Reliability
- RAID
- Redundancy
- RAID 0
- RAID 1



QUIZ

1-What is the minimum number of disks required for RAID1?

- a) 1
- b) 2
- c) 4
- d) 5

ANSWER: b

2-Optical disk technology uses

- a) Helical scanning
- b) DAT
- c) A laser beam
- d) RAID

ANSWER: d



LECTURE -3



- **Disk Array**
- **Data striping**
- **Reliability**
- **RAID**
- **Redundancy**
- **RAID 0 and 1**



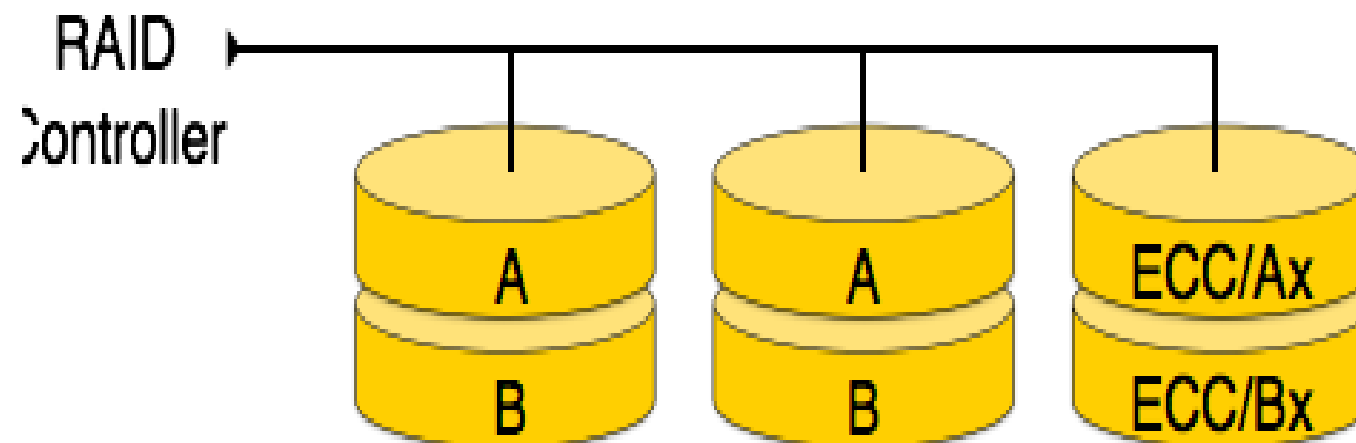
OBJECTIVE

- **RAID-2**
- **Advantages and Disadvantages of RAID 2**
- **RAID-3**
- **Advantages and Disadvantages of RAID 3**
- **RAID-4**

LEVELS OF REDUNDANCY:

Level 2: Error-Correcting Codes

- RAID 2 consists of bit-level striping using hamming code parity. each data bit in a word is recorded on a separate disk and ECC code of data words is stored on different set disks.
- Due to its high cost and complex structure, this level is not commercially used. This same performance can be achieved by RAID 3 at a lower cost.



MEMORY HIERARCHY: RAID

Level 2: Error-Correcting Codes

Advantages of RAID 2:

- Uses one designated drive to store parity.
- It uses the hamming code for error detection.

Disadvantages of RAID 2:

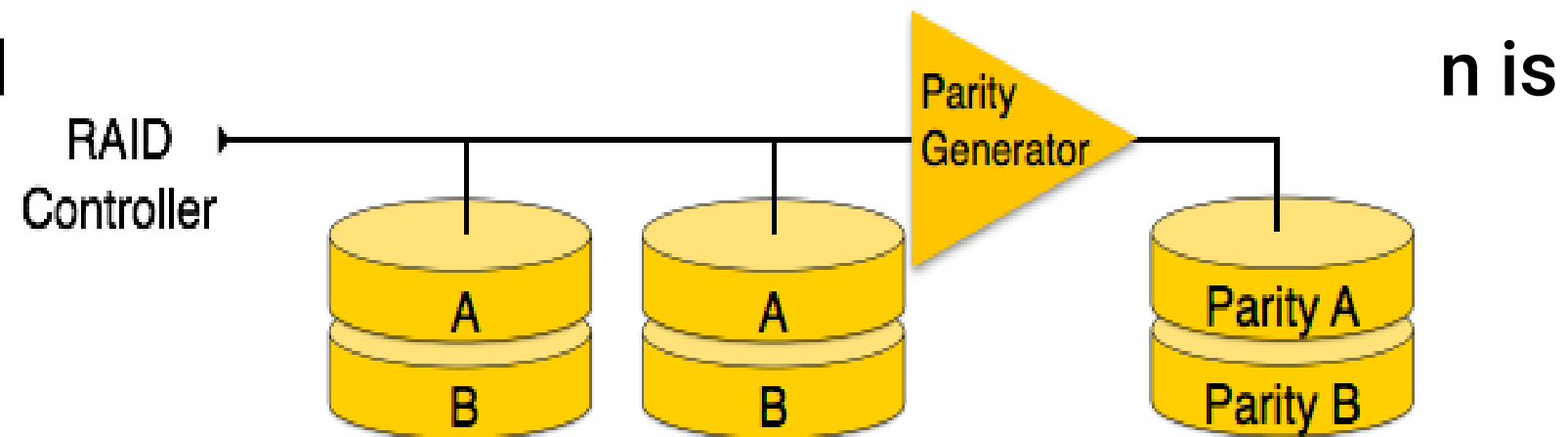
- It requires an additional drive for error detection.



LEVELS OF REDUNDANCY:

Level 3: Bit~Interleaved Parity

- RAID 3 consists of byte-level striping with dedicated parity. the parity information is stored for each disk section and written to a dedicated parity drive.
- In case of drive failure, the parity drive is accessed, and data is reconstructed from the remaining devices. Once the failed drive is replaced, the missing data can be restored on the new drive.
- data can be transferred possible.



LEVELS OF REDUNDANCY:

Level 3: Bit~Interleaved Parity

Advantages of RAID 3:

- Data is regenerated using parity drive.
- It contains high data transfer rates.
- Data is accessed in parallel.

Disadvantages of RAID 3:

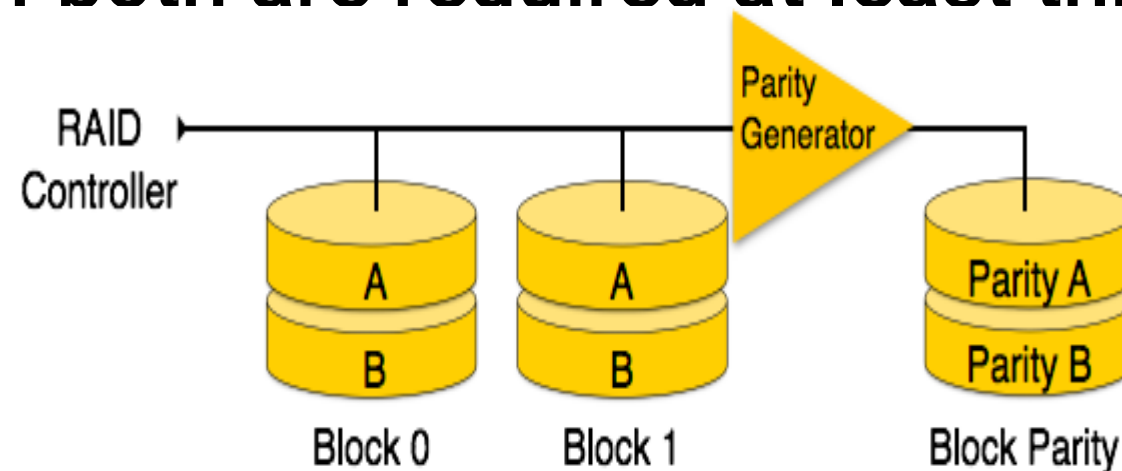
- It required an additional drive for parity.
- It gives a slow performance for operating on small sized files.



LEVELS OF REDUNDANCY:

Level 4: Block Interleaved Parity

- RAID 4 consists of block-level stripping with a parity disk. Instead of duplicating data, the RAID 4 adopts a parity-based approach.
- This level allows recovery of at most 1 disk failure due to the way parity works. if more than one disk fails, then there is no way to recover the data.
- Level 3 and level 4 both are required at least three disks to implement RAID.



SUMMARY

- **RAID-2**
- **Advantages and Disadvantages of RAID 2**
- **RAID-3**
- **Advantages and Disadvantages of RAID 3**
- **RAID-4**



QUIZ

1-Which level of RAID refers to disk mirroring with block striping?

- a) RAID level 1
- b) RAID level 2
- c) RAID level 0
- d) RAID level 3

ANSWER: a

2-Which one of the following is a Stripping technique?

- a) Byte level stripping
- b) Raid level stripping
- c) Disk level stripping
- d) Block level stripping

ANSWER: d



LECTURE -4



- **RAID-2**
- **Advantages and Disadvantages of RAID 2**
- **RAID-3**
- **Advantages and Disadvantages of RAID 3**
- **RAID-4**



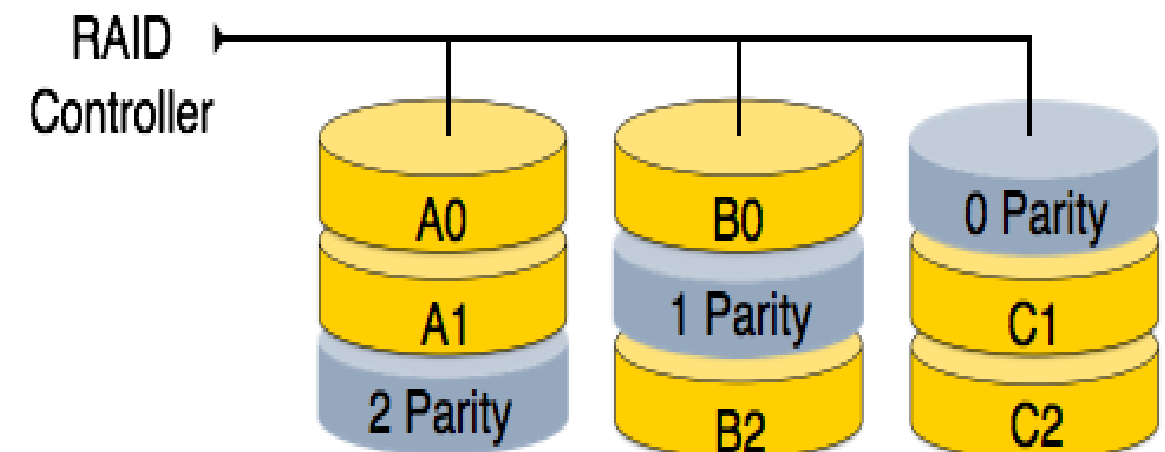
OBJECTIVE

- **RAID-5**
- **Advantages and Disadvantages of RAID 5**
- **RAID-6**
- **Advantages and Disadvantages of RAID 6**
- **RAID-1+0**

LEVELS OF REDUNDANCY:

Level 5: Block-Interleaved Distributed Parity

- RAID 5 is a slight modification of the RAID 4 system. The only difference is that in RAID 5, the parity rotates among the drives.
- It consists of block-level striping with DISTRIBUTED parity.
- Same as RAID 4, this level allows recovery of at most 1 disk failure. If more than one disk fails, then there is no way for data recovery.



LEVELS OF REDUNDANCY:

Level 5: Block-Interleaved Distributed Parity

Advantages of RAID 5:

- Cost effective and provides high performance.
- parity is distributed across the disks in an array.
- It is used to make the random write performance better.

Disadvantages of RAID 5:

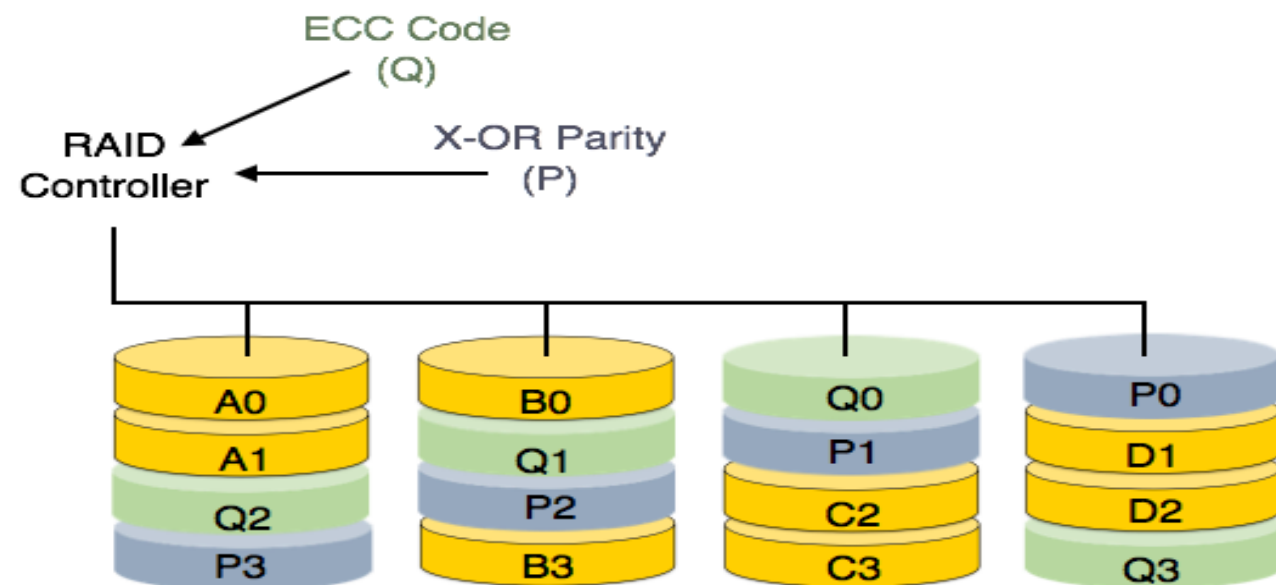
- Disk failure recovery takes longer time as parity has to be calculated from all available drives.
- This level cannot survive in concurrent drive failure.



LEVELS OF REDUNDANCY:

Level 6: P+Q Redundancy

- This level is an extension of RAID 5. It contains block-level stripping with 2 parity bits.
- In RAID 6, you can survive 2 concurrent disk failures. Suppose you are using RAID 5, and RAID 1. When your disks fail, you need to replace the failed disk because if simultaneously another disk fails then you won't be able to recover any of the data, so in this case RAID 6 plays its part where you can survive two concurrent disk failures before



LEVELS OF REDUNDANCY:

Level 6: P+Q Redundancy

Advantages of RAID 6:

- Performs RAID 0 to strip data and RAID 1 to mirror. stripping is performed before mirroring.
- Drives required should be multiple of 2.

Disadvantages of RAID 6:

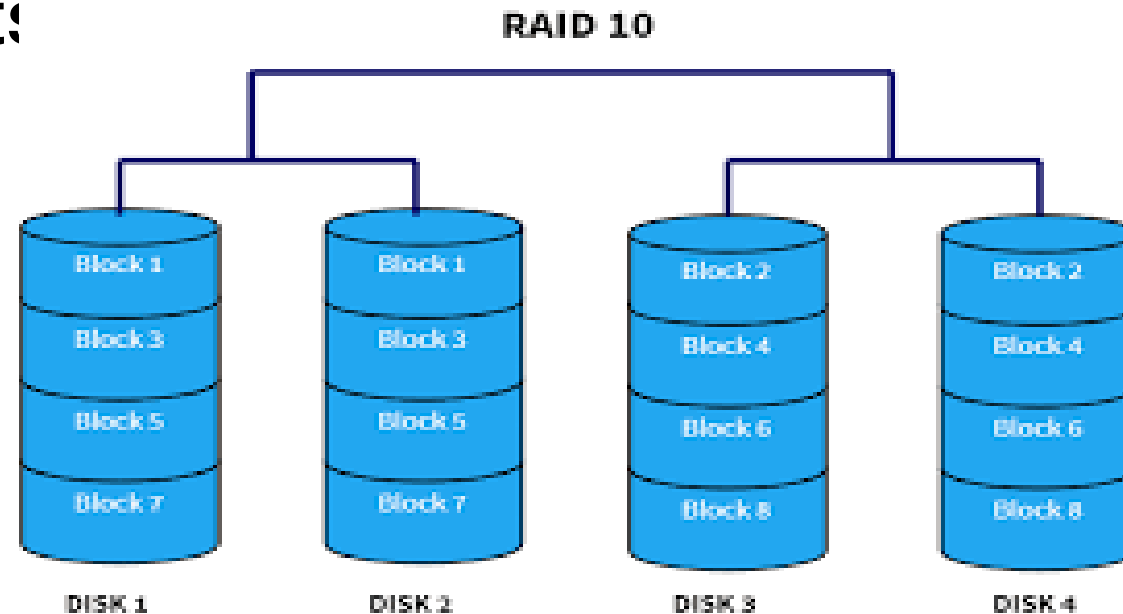
- It is not utilized 100% disk capability as half is used for mirroring.
- It contains very limited scalability.



LEVELS OF REDUNDANCY:

RAID 10 (RAID 1+0)

- RAID 10, also known as RAID 1+0, is a [RAID](#) configuration that combines [disk mirroring](#) and [disk striping](#) to protect data.
- It requires a minimum of four disks and stripes data across mirrored pairs.
- As long as one disk in each mirrored pair is functional, data can be retrieved. If two disks in the same mirrored pair fail, all data will be lost because there is no [parity](#) in the striped sets.



SUMMARY

- **RAID-5**
- **Advantages and Disadvantages of RAID 5**
- **RAID-6**
- **Advantages and Disadvantages of RAID 6**
- **RAID-1+0**



QUIZ

The RAID level which mirroring is done along with stripping is

- a) RAID 1+0
- b) RAID 0
- c) RAID 2
- d) Both RAID 1+0 and RAID 0

Answer:d

Which one of the following is not a secondary storage?

- a) magnetic disks
- b) magnetic tapes
- c) ram
- d) none of the mentioned

Answer:c



LECTURE -5



- **RAID-5**
- **Advantages and Disadvantages of RAID 5**
- **RAID-6**
- **Advantages and Disadvantages of RAID 6**
- **RAID-1+0**



OBJECTIVE

- **DISK SPACE MANAGEMENT**
- **Keeping Track of Free Blocks**
- **OS File Systems to Manage Disk Space**

DISK SPACE MANAGEMENT

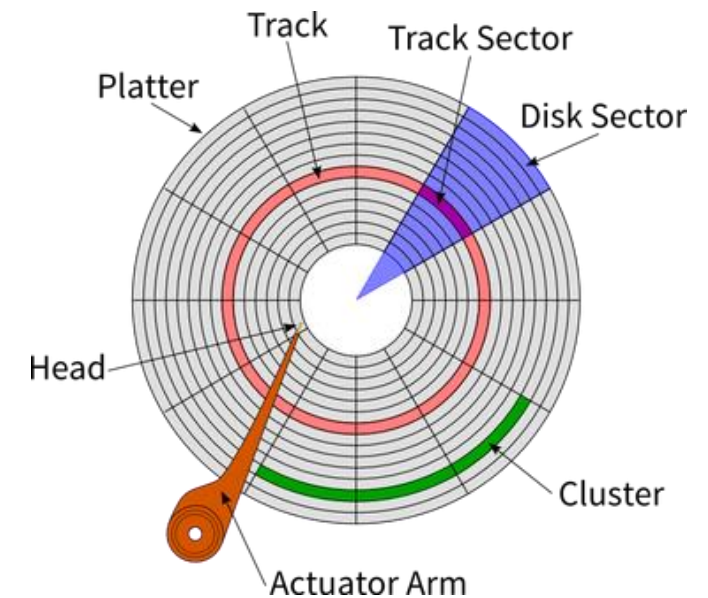
- The lowest level of software in the DBMS architecture called the disk space manager, manages space on disk.
- Abstractly, the disk space manager supports the concept of a page as a unit of data and provides commands to allocate or deallocate a page and read or write a page.



DISK SPACE MANAGEMENT

KEEPING TRACK OF FREE BLOCKS:

- The disk space manager keeps track of which disk blocks are in use.
- blocks are initially allocated sequentially on disk, subsequent allocations and deallocations could in general create 'holes.'
- One way to keep track of block usage is to maintain a. list of free blocks.
- second way is to maintain a bitmap with one bit for each disk block,
which indicates whether a block is in use or not.



DISK SPACE MANAGEMENT

Using OS File Systems to Manage Disk Space

- Operating systems also manage space on disk. Typically, an operating system supports the abstraction of a *file as a sequence of bytes*.
- The OS manages space on the disk and translates requests, such as "Read byte i of file f ," into corresponding low-level instructions: "Read block m of track t of cylinder c of disk d ."
- A database disk space manager could be built using OS files.

SUMMARY

- **DISK SPACE MANAGEMENT**
- **Keeping Track of Free Blocks**
- **OS File Systems to Manage Disk Space**



QUIZ

1-Operating system is responsible for

- a) disk initialization
- b) booting from disk
- c) bad-block recovery
- d) all of the mentioned

Answer: d

2- A magneto-optic disk is :

- a) primary storage
- b) secondary storage
- c) tertiary storage
- d) none of the mentioned

Answer: c



LECTURE -6



- **DISK SPACE MANAGEMENT**
- **Keeping Track of Free Blocks**
- **OS File Systems to Manage Disk Space**

OBJECTIVE

- **Buffer manager**
- **Replacement policy**
- **pin_count and dirty.**
- **Pinning**
- **Unpinning**



BUFFER MANAGER

- Suppose that the database contains 1 million pages, but only 1000 pages of main memory are available for holding.
- Because all the data cannot be brought into main memory at one time, the DBMS must bring pages into main memory as they are needed and, in the process, decide what existing page in main memory to replace to make space for the new page.
- The policy used to decide which page to replace is called **the replacement policy**.



BUFFER MANAGER

- **The buffer manager** is the software layer responsible for bringing pages from disk to main memory as needed.
- The buffer manager manages the available main memory by partitioning it into a collection of pages, which we collectively refer to as the **buffer pool**.
- The main memory pages in the buffer pool are called **frames**



BUFFER MANAGER

- Higher levels of the DBMS code can be written without worrying about whether data pages are in memory or not; they ask the buffer manager for the page, and it is brought into a frame in the buffer pool if it is not already there.
- the higher-level code that requests a page must also release the page when it is no longer needed, by informing the buffer manager, so that the frame containing the page can be reused.
- The higher-level code must also inform the buffer manager if it modifies the requested page; the buffer manager then makes sure that the change is propagated to the copy of the page on disk.



BUFFER MANAGER

- The buffer manager maintains some book keeping information and two variables for each frame in the pool: *pin_count* and *dirty*.
- The number of times that the page currently in a given frame has been requested but not released-the number of current users of the page is recorded in the *pin_count* variable for that frame.
- The Boolean variable *dirty* indicates whether the page has been modified since it was brought into the buffer pool from disk.



BUFFER MANAGER

Initially, the *pin_count* for every frame is set to 0, and the *dirty* bits are turned off. When a page is requested the buffer manager does the following:

1. Checks the buffer pool to see if some frame contains the requested page and, if so, increments the *pin_count* of that frame. If the page is not in the pool, the buffer manager brings it in as follows:

(a) Chooses a frame for replacement, using the replacement policy, and increments its *pin_count*.

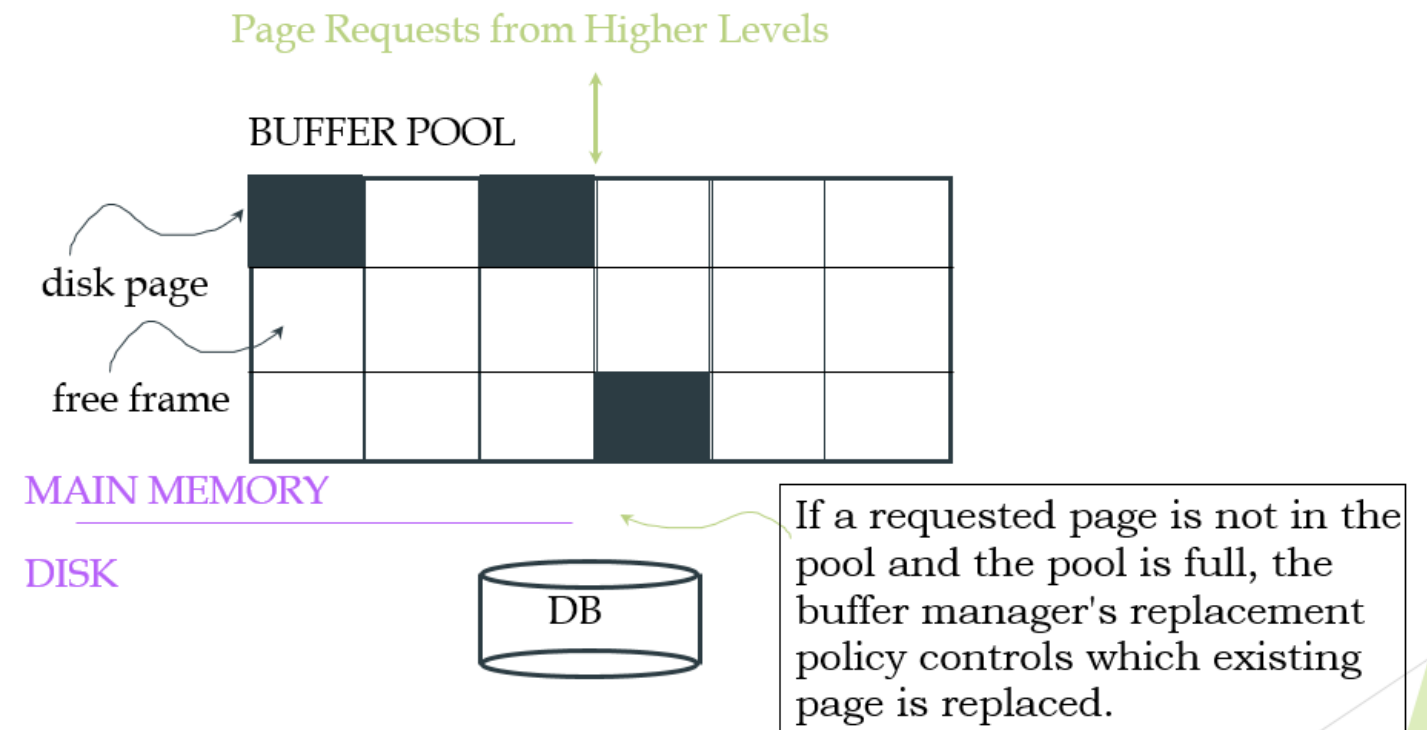
(b) If the *dirty* bit for the replacement frame is on, writes the page it contains to disk (that is, the disk copy of the page is overwritten with the contents of the frame).

 (c) Reads the requested page into the replacement frame.

BUFFER MANAGER

2.Returns the (main memory) address of the frame containing the requested page to the requestor.

- Incrementing *pin_count* is often called **pinning**.
- the *pin_count* of the frame containing the requested page is decremented. This is called **unpinning the page**.



SUMMARY

- **Buffer manager**
- **Replacement policy**
- **pin_count and dirty.**
- **Pinning**
- **Unpinning**



QUIZ

1-Which of the following is the oldest database model?

- a) Relational**
- b) Hierarchical**
- c) Physical**
- d) Network**

Answer: d

2-Which one of the following is not a secondary storage?

- a) magnetic disks**
- b) magnetic tapes**
- c) ram**
- d) none of the mentioned**

Answer:c



LECTURE -7



- **Buffer manager**
- **Replacement policy**
- **pin_count and dirty.**
- **Pinning**
- **Unpinning**

OBJECTIVE

- **Buffer Replacement Policies**
- **sequential flooding**
- **FILES OF RECORDS**
- **Implementing Heap Files**
- **Linked List of Pages**
- **Directory of Pages**



BUFFER REPLACEMENT POLICIES

- The best-known replacement policy is least recently used (LRU). This can be implemented in the buffer manager using a queue of pointers to frames with *pin_count* 0.
- The page chosen for replacement is the one in the frame at the head of the queue.
- A variant of LRU, called clock replacement, has similar behavior but less overhead.
- Using LRU, every scan of the file will result in reading every page of the file! In this situation, called **sequential flooding**, LRU is the *worst* possible replacement strategy.
- Other replacement policies include **first in first out (FIFO)** and **most recently used (MRU)**, which also entail overhead similar to LRU, and random, among others.



FILES OF RECORDS

- the way pages are stored on disk and brought into main memory, to the way pages are used to store records and organized into logical collections or *files*.
- Higher levels of the DBMS code treat a page as effectively being a collection of records, ignoring the representation and storage details.



FILES OF RECORDS

Implementing Heap Files

- The data in the pages of a heap file is not ordered in any way, and the only guarantee is that one can retrieve all records in the file by repeated requests for the next record.
- Supported operations on a heap file include *Create* and *destroy* files, *insert* a record, *delete* a record with a given rid, *get* a record with a given rid, and *scan* all records in the file.
- We must keep track of the pages in each heap file to support scans, and we must keep track of pages that contain free space to implement insertion efficiently

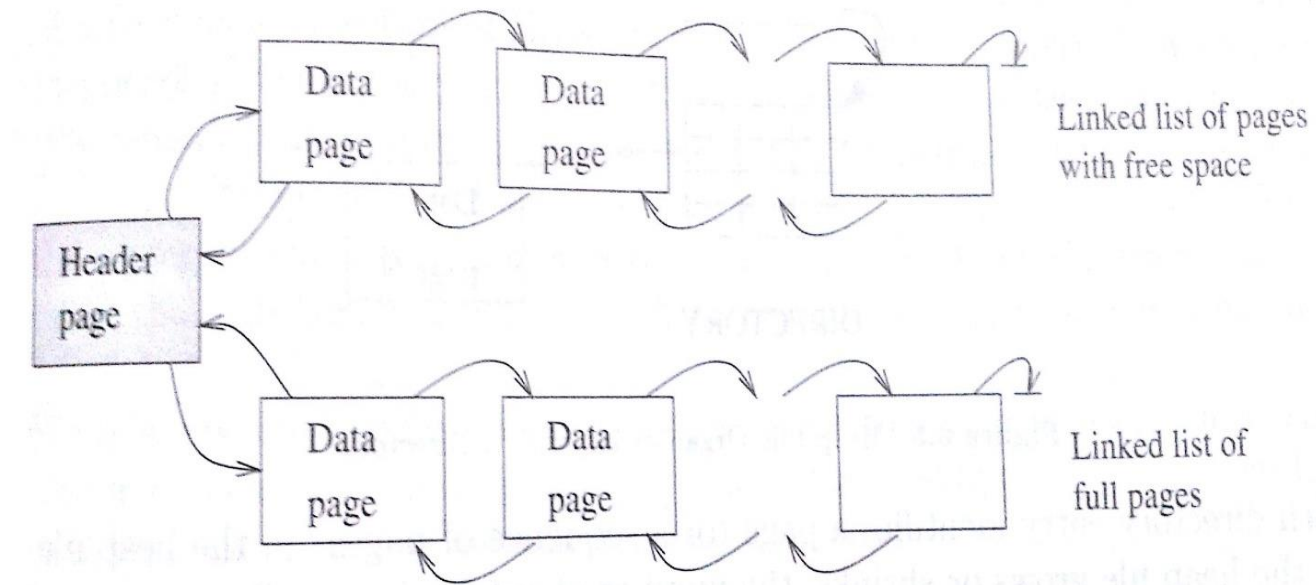
FILES OF RECORDS

Implementing Heap Files

- There are two alternative ways to maintain this information.

1. Linked List of Pages:

- ❑ One possibility is to maintain a heap file as a doubly linked list of pages.
- ❑ The DBMS can remember where the first page is located by maintaining a table containing pairs of *(heap_file_name, page_Laddr)* in a known location on disk. We call the first page of the file the *header page*.



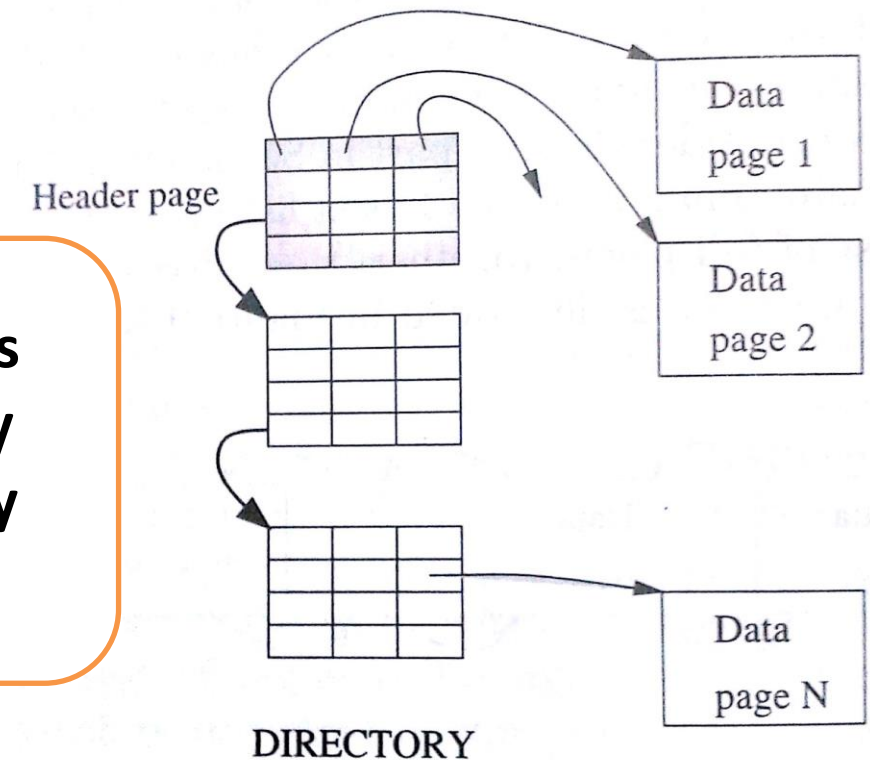
FILES OF RECORDS

Implementing Heap Files

2. Directory of Pages

- An alternative to a linked list of pages is to maintain a **directory of pages**.
- Each directory entry identifies a page (or a sequence of pages) in the heap file.

Free space is managed by Bit per entry count.



SUMMARY

- **Buffer Replacement Policies**
- **sequential flooding**
- **Files of records**
- **Implementing Heap Files**
- **Linked List of Pages**
- **Directory of Pages**



QUIZ

1-The file organization which allows us to read records that would satisfy the join condition by using one block read is

- a) Heap file organization
- b) Sequential file organization
- c) Clustering file organization
- d) Hash file organization

Answer:c

2-Large collection of files are called

- a) Fields
- b) Records
- c) Database
- d) Sectors

Answer:c



LECTURE -8



- **Buffer Replacement Policies**
- **sequential flooding**
- **Files of records**
- **Implementing Heap Files**
- **Linked List of Pages**
- **Directory of Pages**



OBJECTIVE

- **Record Arrangement**
- **fixed length records**
- **fixed length records Problem and solution**
- **Variable length Records**
- **Variable length Records Problem and solution**

PAGE FORMATS

How collection of records are arranged in a page?

Think a page as collection of records

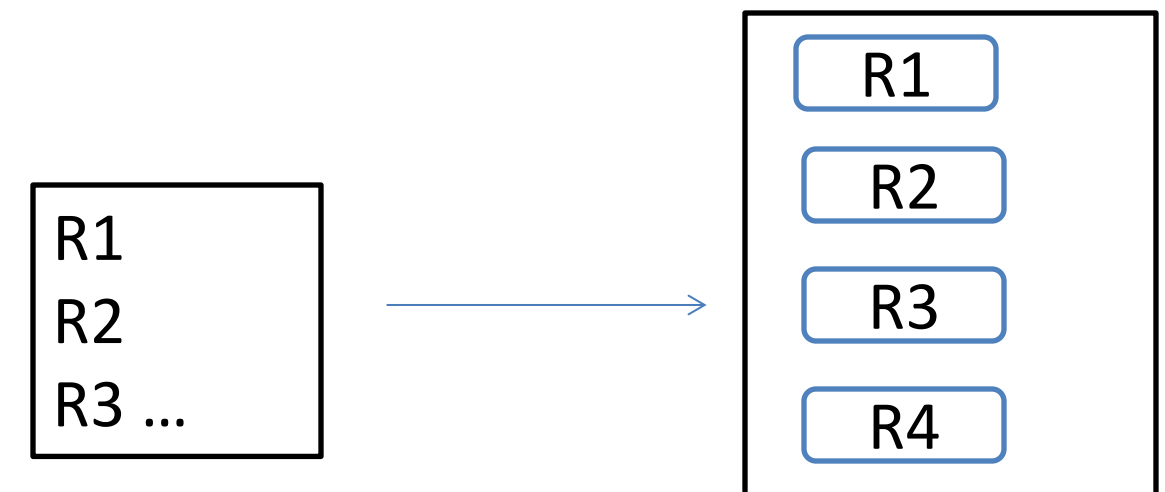
- A record in a page is identified by using the pair (pageID, slotNo) as RID.

➤ Types of Records?

1) fixed length records

1) fixed length records

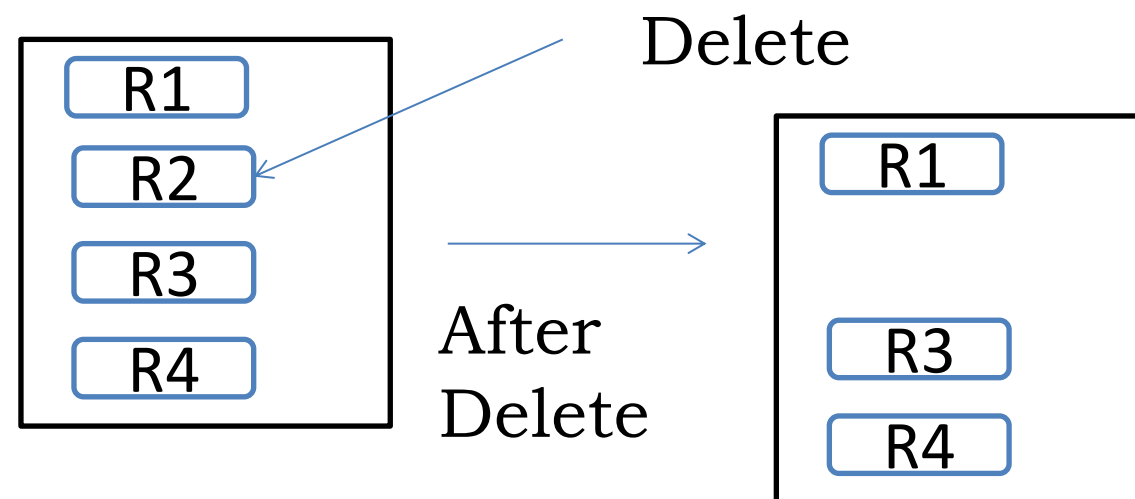
- if all records on the page are guaranteed to be of same length. Then records can be uniformly arranged in a page.



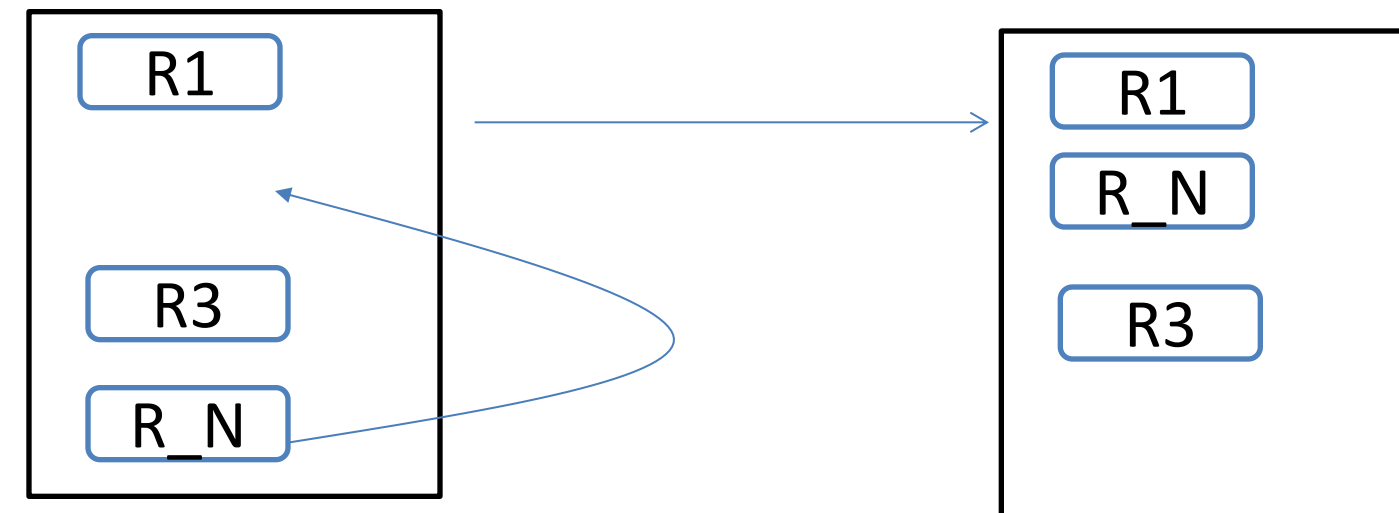
PAGE FORMATS

fixed length records

ISSUES: how to keep track of empty slots if a record is deleted as shown below.



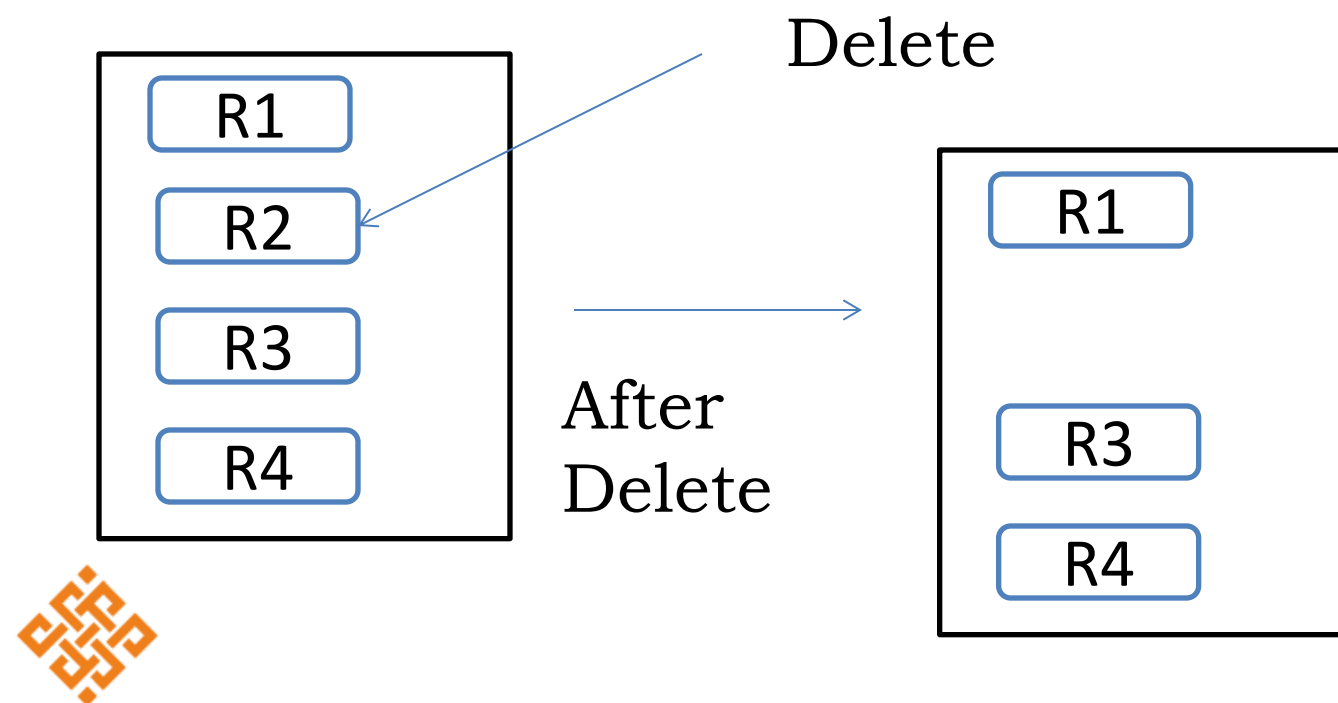
Soln: Alternative-1 store records in sequence of first to N. if a record is deleted replace the empty space with Nth record. As shown in Fig.



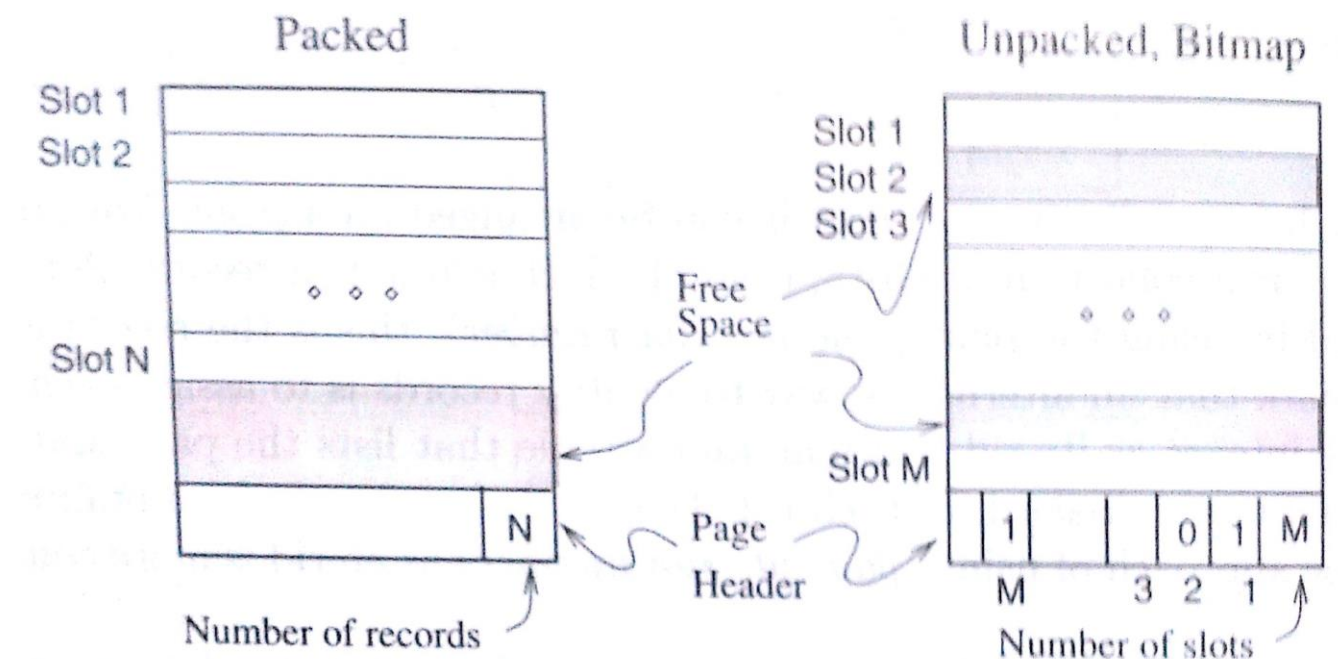
PAGE FORMATS

fixed length records

ISSUES: how to keep track of empty slots if a record is deleted as shown below.



Soln: Alternative-2 handle deletion using array of bits. One per slot to keep track of free slots.



PAGE FORMATS

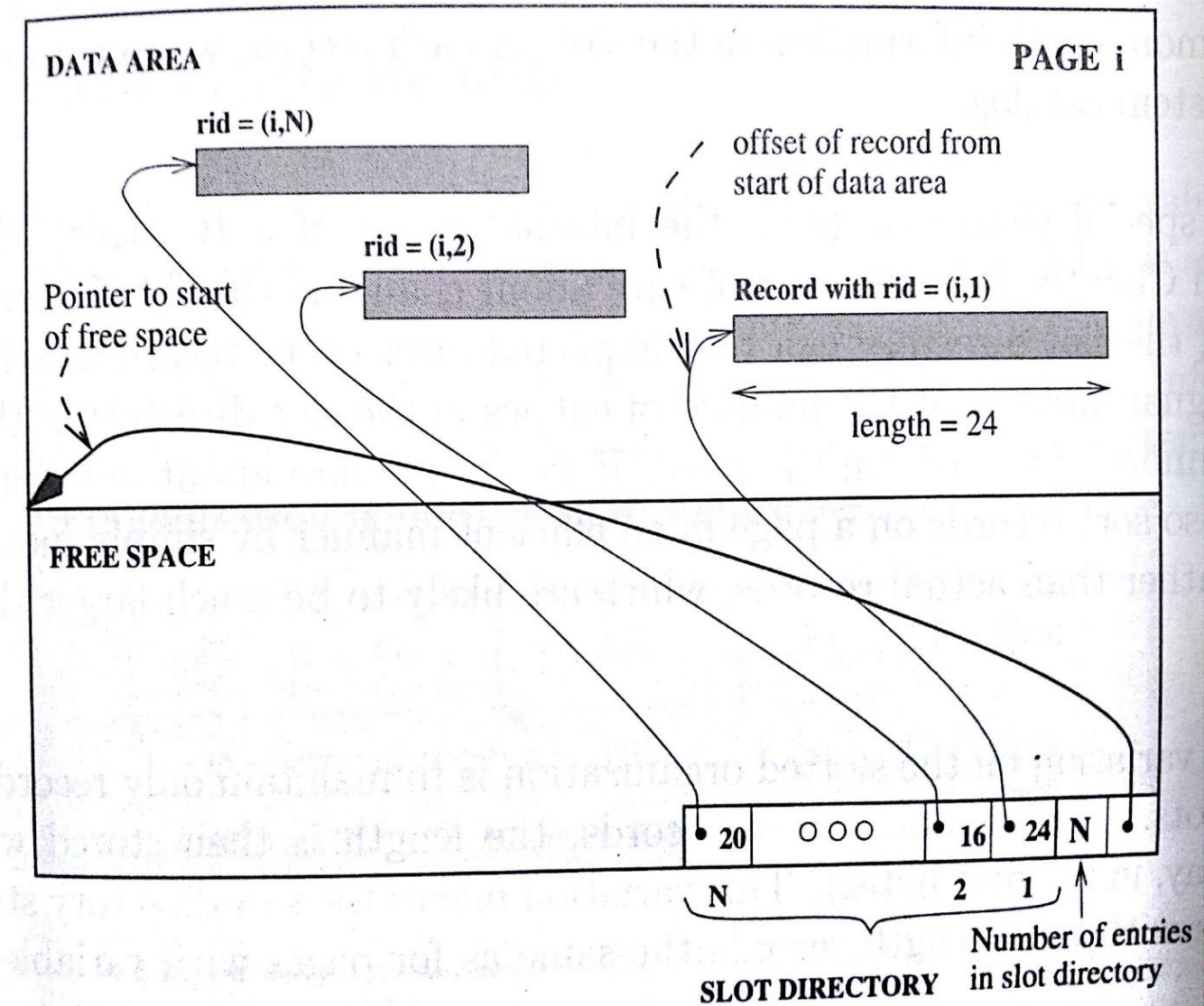
Variable length Records

- If records are variable length then we cannot divide them in to equal slots.
- To insert new record → find empty space of exact size. Other wise space is wasted.

what is the solution

Maintain a directory of slots for each page with a

<record-offset record length> pair



SUMMARY

- **Record Arrangement**
- **fixed length records**
- **fixed length records Problem and solution**
- **Variable length Records**
- **Variable length Records Problem and solution**



QUIZ

1-Storing a separate copy of the database at multiple locations is ?

- A) Data Replication**
- B) Horizontal Partitioning**
- C) Vertical Partitioning**
- D) Horizontal and Vertical Partitioning**

Answer: A

2-A unit of storage that can store one or more records in a hash file organization is denoted as

- a) Buckets**
- b) Disk pages**
- c) Blocks**
- d) Nodes**

Answer:a



LECTURE -9



- **Record Arrangement**
- **fixed length records**
- **fixed length records Problem and solution**
- **Variable length Records**
- **Variable length Records Problem and solution**



OBJECTIVE

- **Index**
- **Primary Index**
- **Secondary Index**
- **Clustering Index**

STRUCTURED INDEXING

- We know that data is stored in the form of records. Every record has a key field, which helps it to be recognized uniquely.
- Indexing is a data structure technique to efficiently retrieve records from the database files based on some attributes on which the indexing has been done.
- Indexing is defined based on its indexing attributes. Indexing can be of the following types



STRUCTURED INDEXING

1. Primary Index

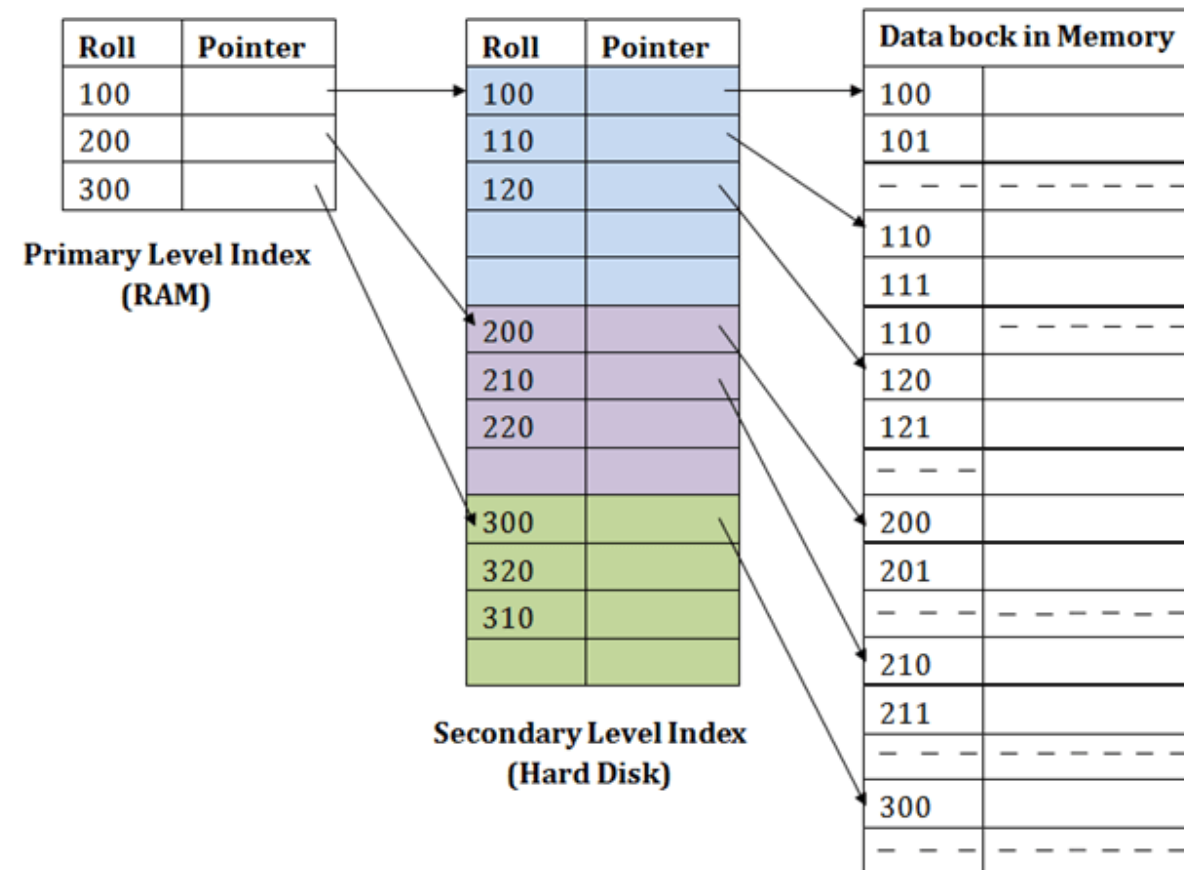
- Primary index is defined on an ordered data file. The data file is ordered on a **key field**.
- The key field is generally the primary key of the relation.

UP	•	→	UP	Agra	1,604,300
USA	•	→	USA	Chicago	2,789,378
Nepal	•	→	Nepal	Kathmandu	1,456,634
UK	•	→	UK	Cambridge	1,360,364



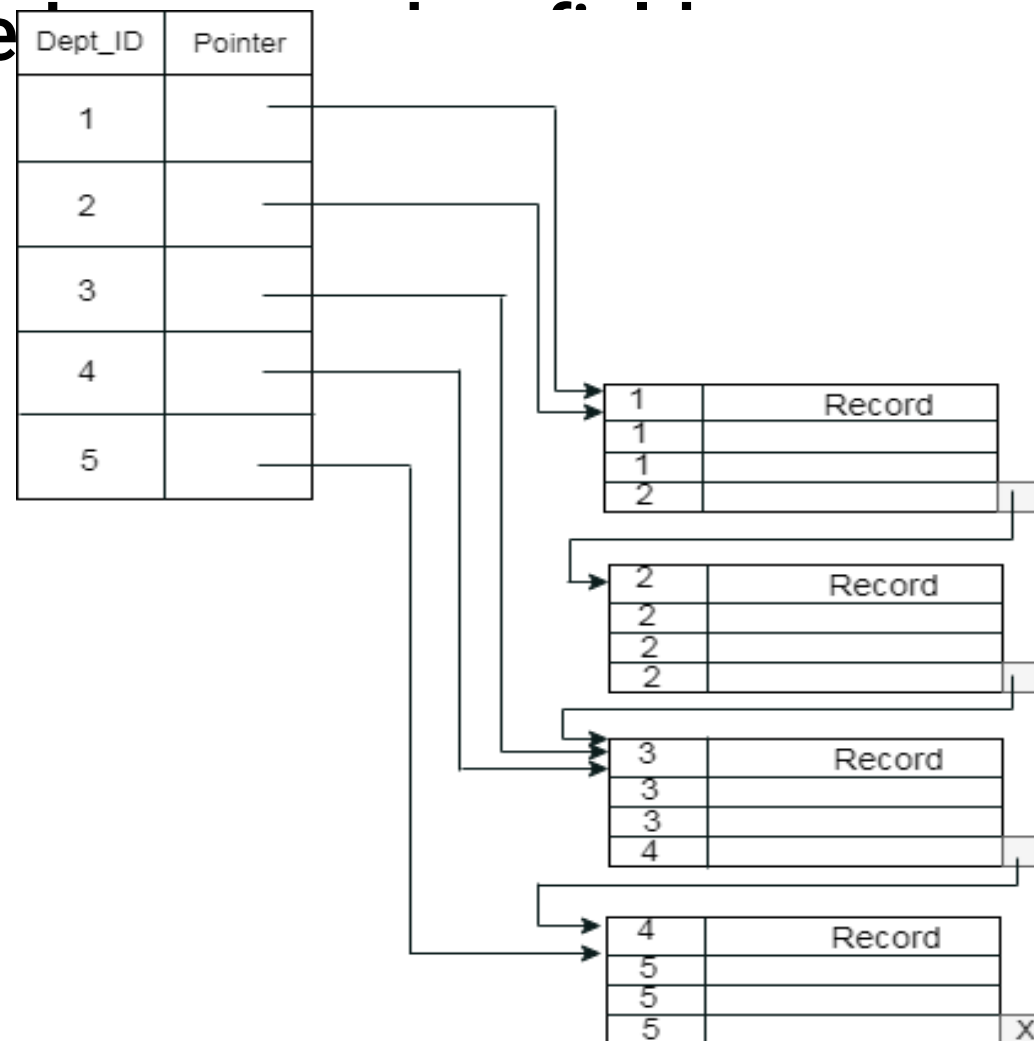
STRUCTURED INDEXING

2- Secondary Index – Secondary index may be generated from a field which is a candidate key and has a unique value in every record, or a non-key with duplicate values.



STRUCTURED INDEXING

3. Clustering Index – Clustering index is defined on an ordered data file. The data file is ordered



SUMMARY

- **Index**
- **Primary Index**
- **Secondary Index**
- **Clustering Index**



QUIZ

1-Does index take space in the disk?

- a) It stores memory as and when required
- b) Yes, Indexes are stored on disk
- c) Indexes are never stored on disk
- d) Indexes take no space

Answer:b

2-Which string function returns the index of the first occurrence of substring?

- a) INSERT()
- b) INSTR()
- c) INSTRING()
- d) INFSTR()

Answer:b



LECTURE -10



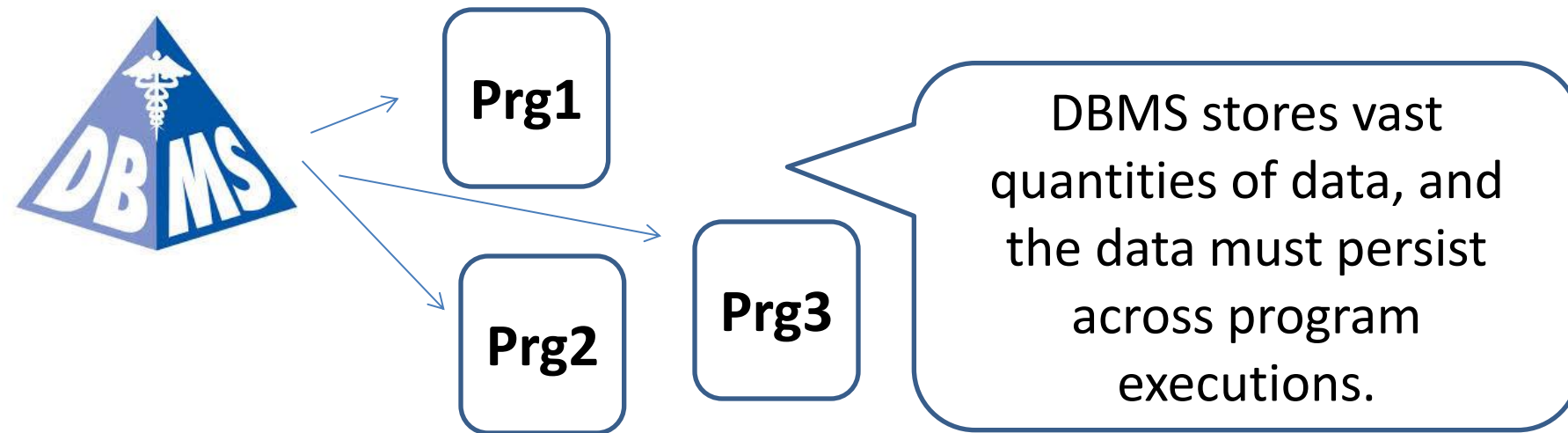
- **Index**
- **Primary Index**
- **Secondary Index**
- **Clustering Index**

OBJECTIVE

- **Data on External Storage**
- **Magnetic tapes**
- **Page**
- **cost of page I/O**



DATA ON EXTERNAL STORAGE



Therefore, data is stored on external storage devices such as **disks and tapes**. And fetched into main memory as needed for processing.



DATA ON EXTERNAL STORAGE



POINTS TO REMEMBER

➤ Disks:

- Can retrieve random page at fixed cost
- But reading several consecutive pages is much cheaper than reading them in random order.



➤ Tapes (magnetic tapes):

- Can only read pages in sequence
- Cheaper than disks, used for archival storage.
(Archival storage: data that may not be actively needed)



DATA ON EXTERNAL STORAGE



POINTS TO REMEMBER

➤ Page:

- ❑ The unit of information read from or written to disk is a page.
- ❑ The size of a page is a DBMS parameter, and typical values are 4KB or 8KB.

➤ Cost of page I/O

The cost of page I/O (input from disk to main memory and output from memory

to disk) dominates the cost of typical database operations, and database systems are carefully optimized to minimize this cost.



SUMMARY

- **Data on External Storage**
- **Magnetic tapes**
- **Page**
- **cost of page I/O**



QUIZ

1-Which of the following is a Disadvantages of replication?

- A) Reduced network traffic
- B) If the database fails at one site, a copy can be located at another site.
- C) Each site must have the same storage capacity.
- D) Each transaction may proceed without coordination across the network.

 **Answer: C**

2-A distributed database can use which of the following strategies?

- A) Totally centralized at one location and accessed by many sites
- B) Partially or totally replicated across sites
- C) Partitioned into segments at different sites
- D) All of the above

Answer: D

LECTURE -11



- **Data on External Storage**
- **Magnetic tapes**
- **Page**
- **cost of page I/O**

OBJECTIVE

- **File**
- **File organization**
- **Index data structures**
- **Search key**
- **Data Reference**
- **Type of Index data structures**



FILE ORGANIZATION

- The **File** is a collection of records. Using the primary key, we can access the records. The type and frequency of access can be determined by the type of file organization which was used for a given set of records.



FILE ORGANIZATION

- **File organization** is a logical relationship among various records. This method defines how file records are mapped onto disk blocks.
- File organization is used to describe the way in which the records are stored in terms of blocks, and the blocks are placed on the storage medium.



FILE ORGANIZATION

- The first approach to map the database to the file is to use the several files and store only one fixed length record in any given file.
- An alternative approach is to structure our files so that we can contain multiple lengths for records.
- Files of fixed length records are easier to implement than the files of variable length records.



INDEX DATA STRUCTURES

- Indexing is a way to optimize the performance of a database by minimizing the number of disk accesses required when a query is processed.
- It is a data structure technique which is used to quickly locate and access the data in a database.

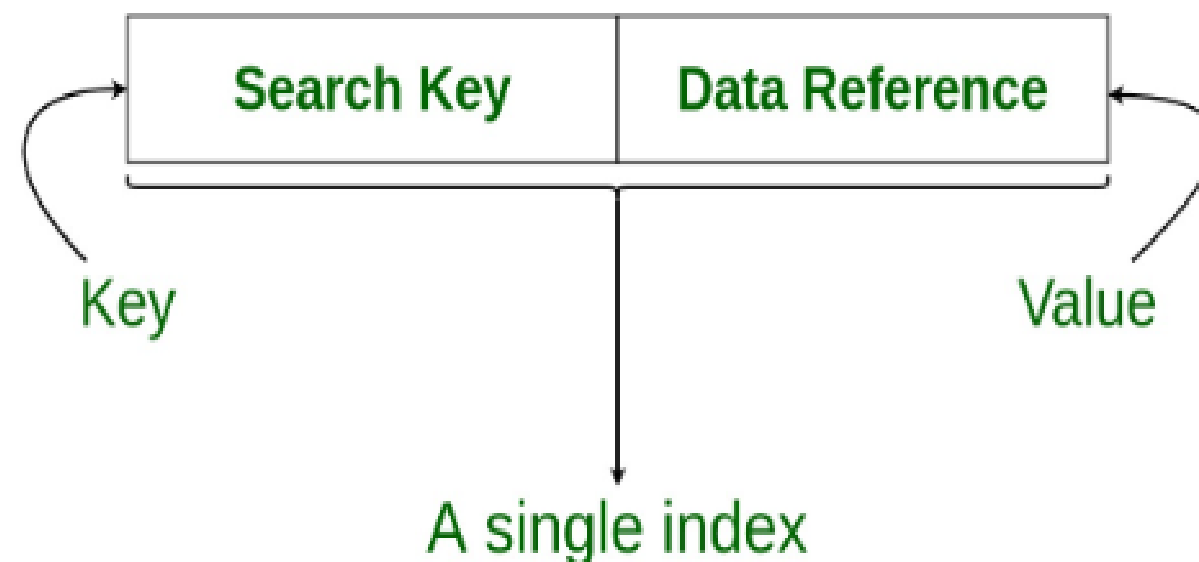


INDEX DATA STRUCTURES

1- The first column is the **Search key** that contains a copy of the primary key or candidate key of the table. These values are stored in sorted order so that the corresponding data can be accessed quickly.

Note: The data may or may not be stored in sorted order.

2- The second column is the **Data Reference** or **Pointer** which contains a set of pointer
particular key v k where that



INDEX DATA STRUCTURES

Two types of Index Data Structures:

- 1) Hash based Indexing
- 2) Tree Based Indexing



SUMMARY

- **File**
- **File organization**
- **Index data structures**
- **Search key**
- **Data Reference**
- **Type of Index data structures**



QUIZ

The file organization which allows us to read records that would satisfy the join condition by using one block read is

- a) Heap file organization
- b) Sequential file organization
- c) Clustering file organization
- d) Hash file organization

Answer: c

Key value pairs is usually seen in

- a) Hash tables
- b) Heaps
- c) Both Hash tables and Heaps
- d) Skip list

Answer: a



LECTURE -12



- **File**
- **File organization**
- **Index data structures**
- **Search key**
- **Data Reference**
- **Type of Index data structures**

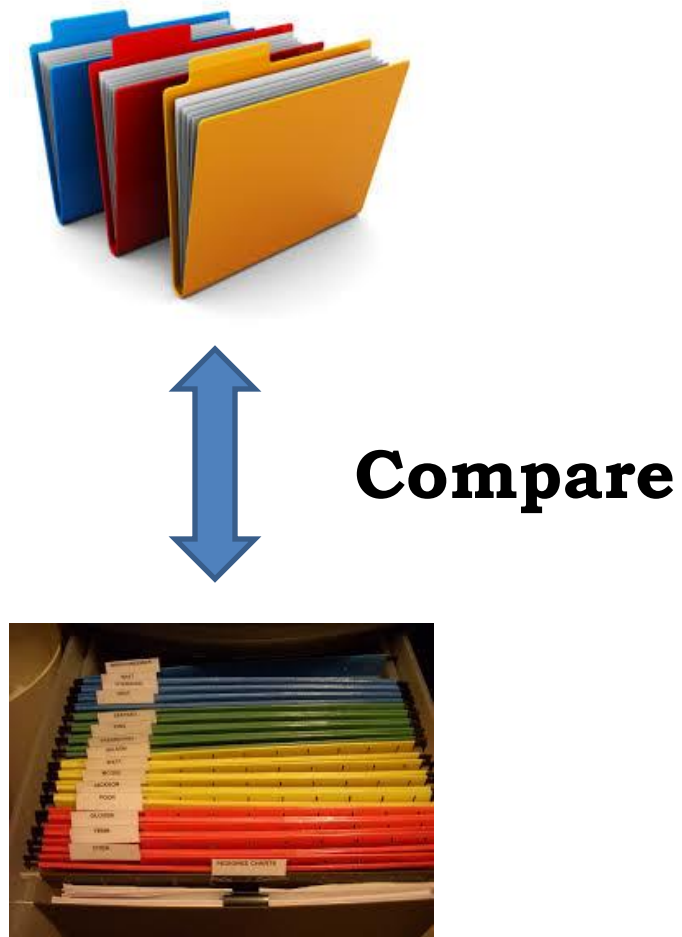


OBJECTIVE

- **Comparison of file organizations**
- **heap file**
- **Clustered B+ tree**
- **Heap file with an unclustered B+ tree**
- **Heap file with an unclustered hash**

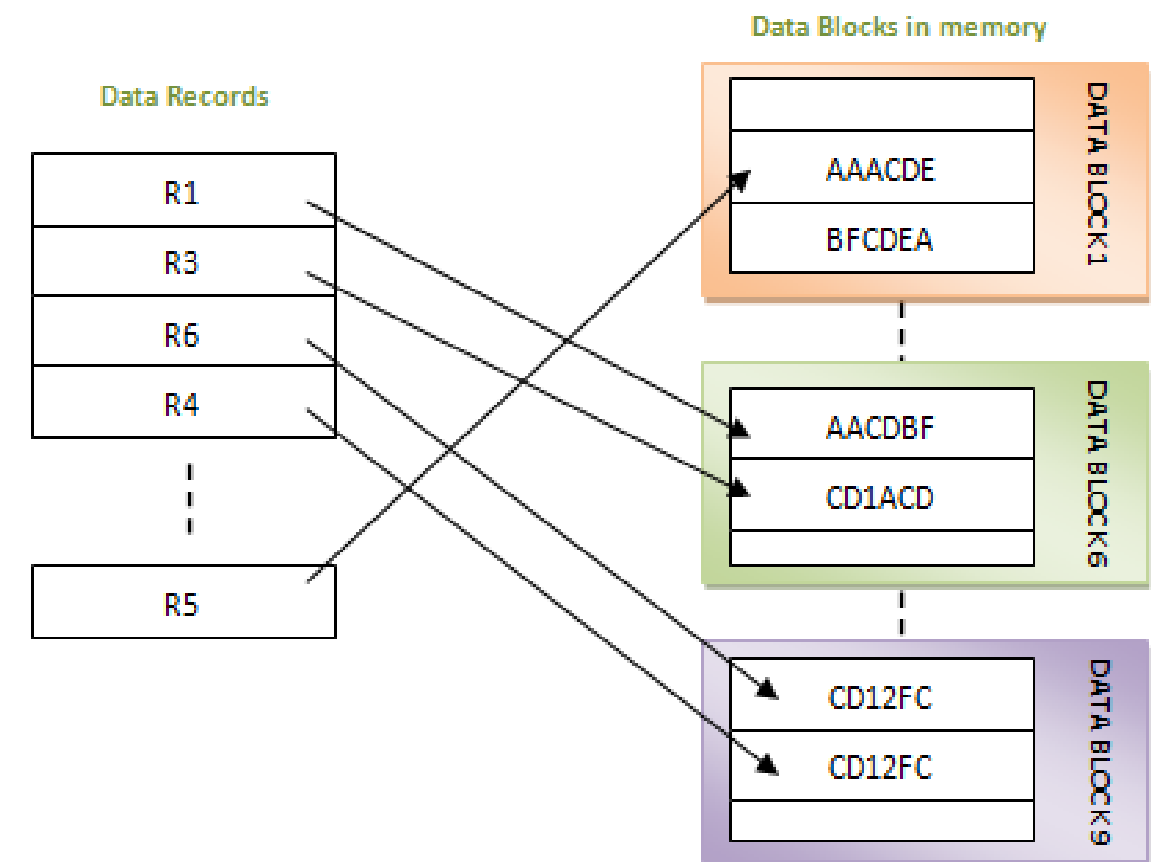
COMPARISON OF FILE ORGANIZATIONS

- The files and indexes are organized according to the composite search key (age, sal) and
- all selection operations are specified on these fields.



COMPARISON OF FILE ORGANIZATIONS

- Different File organizations are:
 1. File of randomly ordered employee records, or **heap file**.

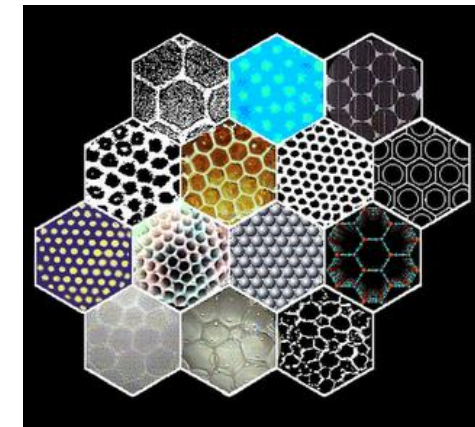


COMPARISON OF FILE ORGANIZATIONS

- **Different File organizations are:**

2. File of employee records sorted on (age, sal).

3. Clustered B+ tree file with search key (age, sal).

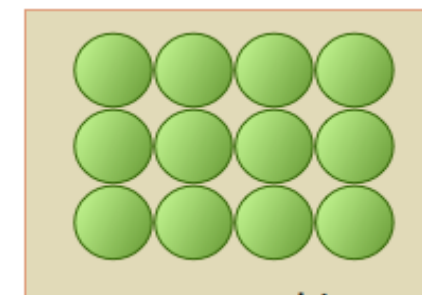
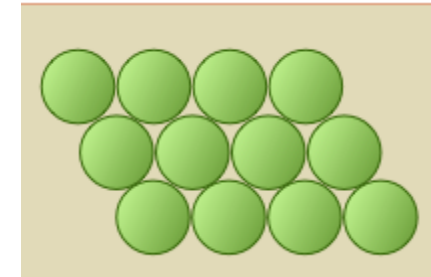


COMPARISON OF FILE ORGANIZATIONS

- **Different File organizations are:**

4. Heap file with an unclustered B+ tree index on (age, sal).

5. Heap file with an unclustered hash index on (age, sal).



SUMMARY

- **Comparison of file organizations**
- **heap file**
- **Clustered B+ tree**
- **Heap file with an unclustered B+ tree**
- **Heap file with an unclustered hash**



QUIZ

1-A transaction manager is ?

- A) Maintains a log of transactions
- B) Maintains before and after database images
- C) Maintains appropriate concurrency control
- D) All of the above.

Answer: D

2-A distributed database has which of the following Advantages over a centralized database?

- A) Software cost
- B) Software complexity
- C) Slow Response
- D) Modular growth

Answer: D



LECTURE -13



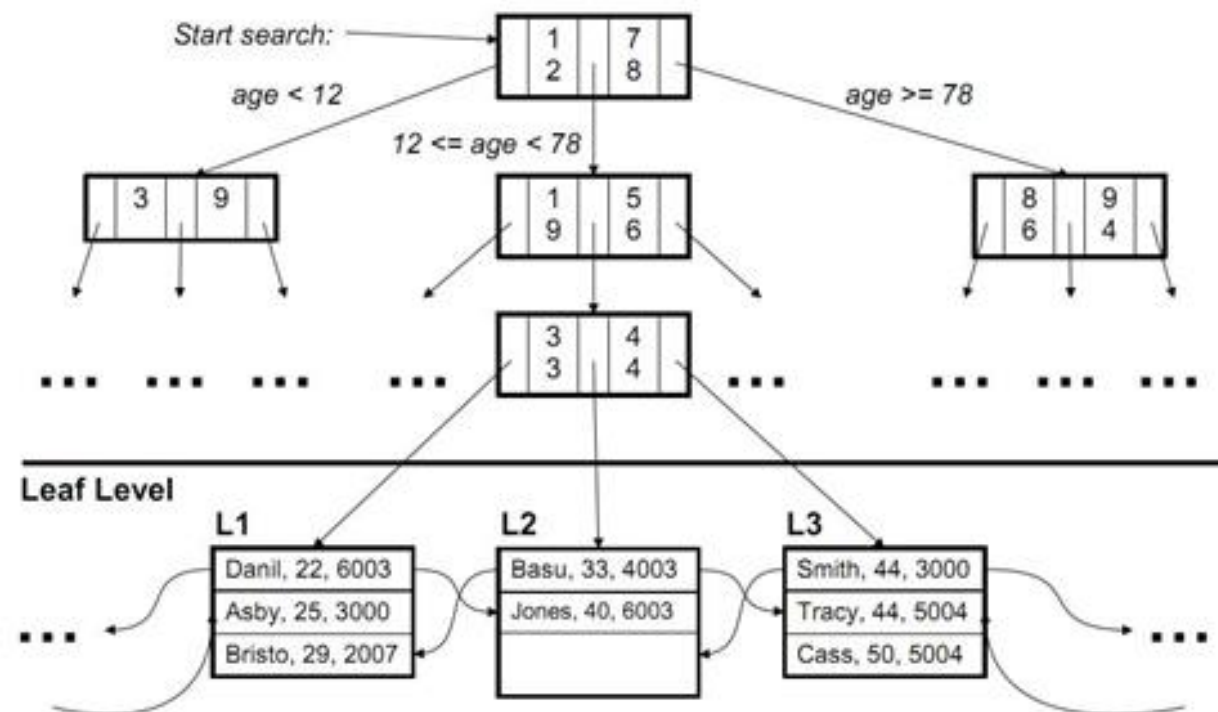
- **Comparison of file organizations**
- **heap file**
- **Clustered B+ tree**
- **Heap file with an unclustered B+ tree**
-  ➤ **Heap file with an unclustered hash**

OBJECTIVE

- **Tree Structured index**
- **Tree Structured index**
- **ISAM (indexed sequential access method)**
- **B+ Trees**

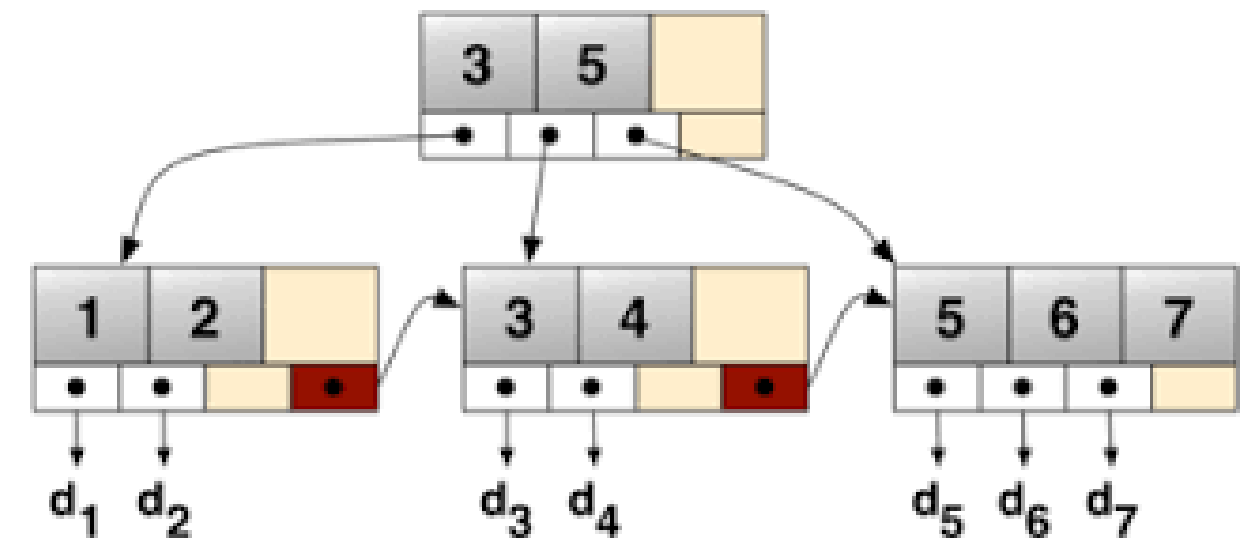
TREE-BASED INDEXING:

- The data entries are arranged in sorted order by search key value and a hierarchical search data structure is maintained.



TREE-BASED INDEXING:

- Tree Structured index
- index storage techniques uses 3 alternatives for data entries k^* :
 - Data record with key value k
 - $\langle k, \text{rid of data record with search key value } k \rangle$
 - $\langle k, \text{list of rids of data records with search key } k \rangle$
- Tree-structured indexing techniques support both range searches and equality searches.



TREE-BASED INDEXING:

Tree Structured index

□ Two techniques available in tree structured indexing:

1. ISAM (indexed sequential access method)

2. B+ Trees

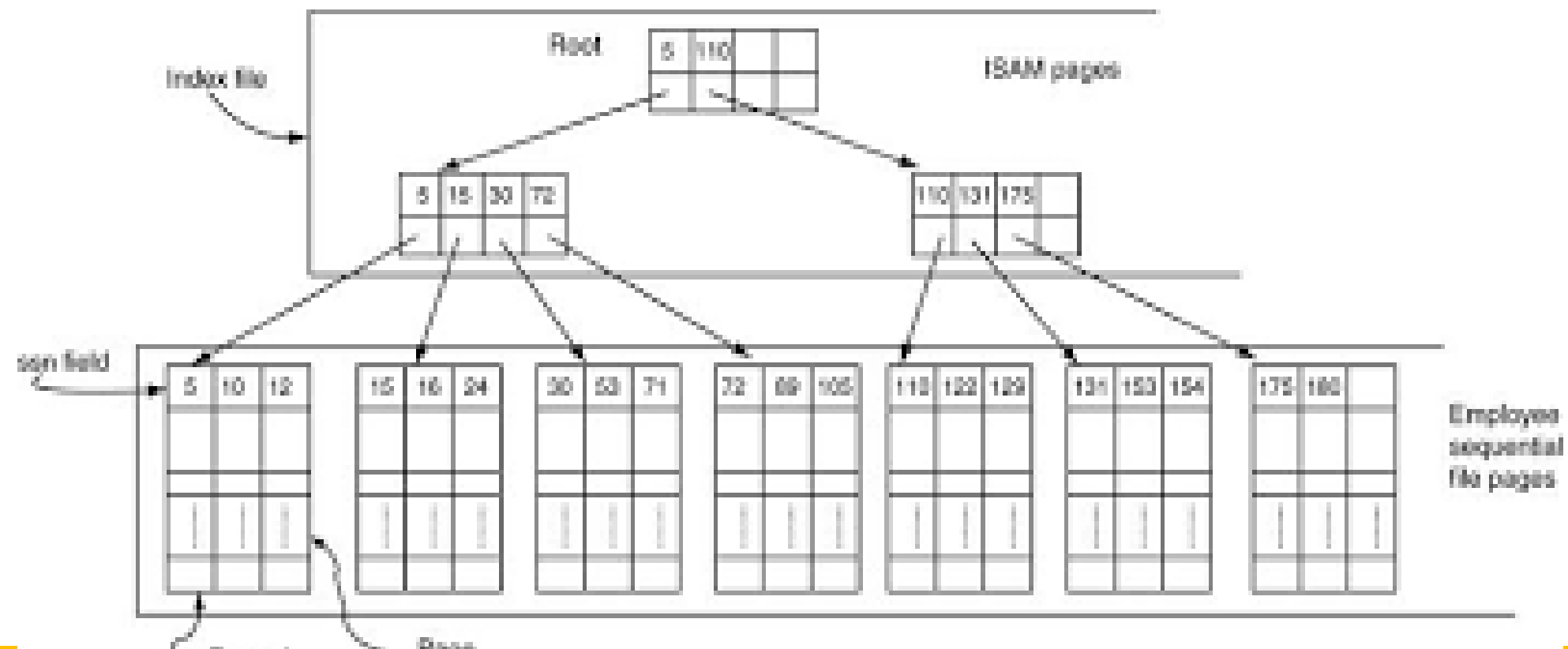
Both supports effective range searches



TREE-BASED INDEXING:

ISAM →

- it is static index structure that is effective when the file is not frequently updated.
- This method is not suitable for a file that grows and shrinks a lot.

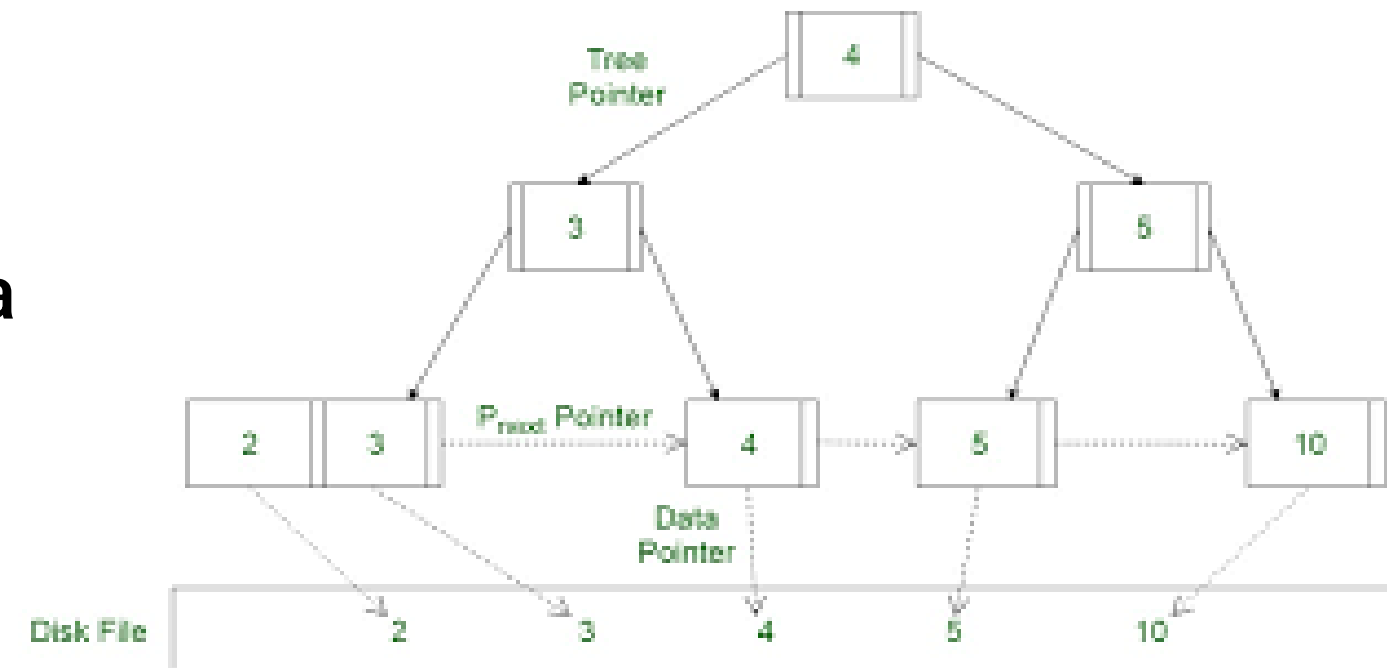


TREE-BASED INDEXING:

B + Trees → A dynamic structure that adjusts to changes in the file gracefully.

Most widely used index structure.

- ✓ because it adjusts well to changes
- ✓ Supports equality search and range sea



SUMMARY

- **Tree Structured index**
- **Tree Structured index**
- **ISAM (indexed sequential access method)**
- **B+ Trees**



QUIZ

1-Which of the reasons will force you to use XML data model in SQL Server ?

- A) Your data is sparse or you do not know the structure of the data**
- B) Your data represents containment hierarchy**
- C) Order is inherent in your data**
- D) All of the Mentioned**



Answer: D

2-Which of the following is not a XML storage option ?

- A) Native storage as XML data type**
- B) Mapping between XML and relational storage**
- C) Small object storage**
- D) None of the Mentioned**

Answer: C

LECTURE -14



- **Tree Structured index**
- **Tree Structured index**
- **ISAM (indexed sequential access method)**
- **B+ Trees**

OBJECTIVE

- **ISAM**
- **ISAM structure**
- **leaf pages**
- **Non leaf pages**



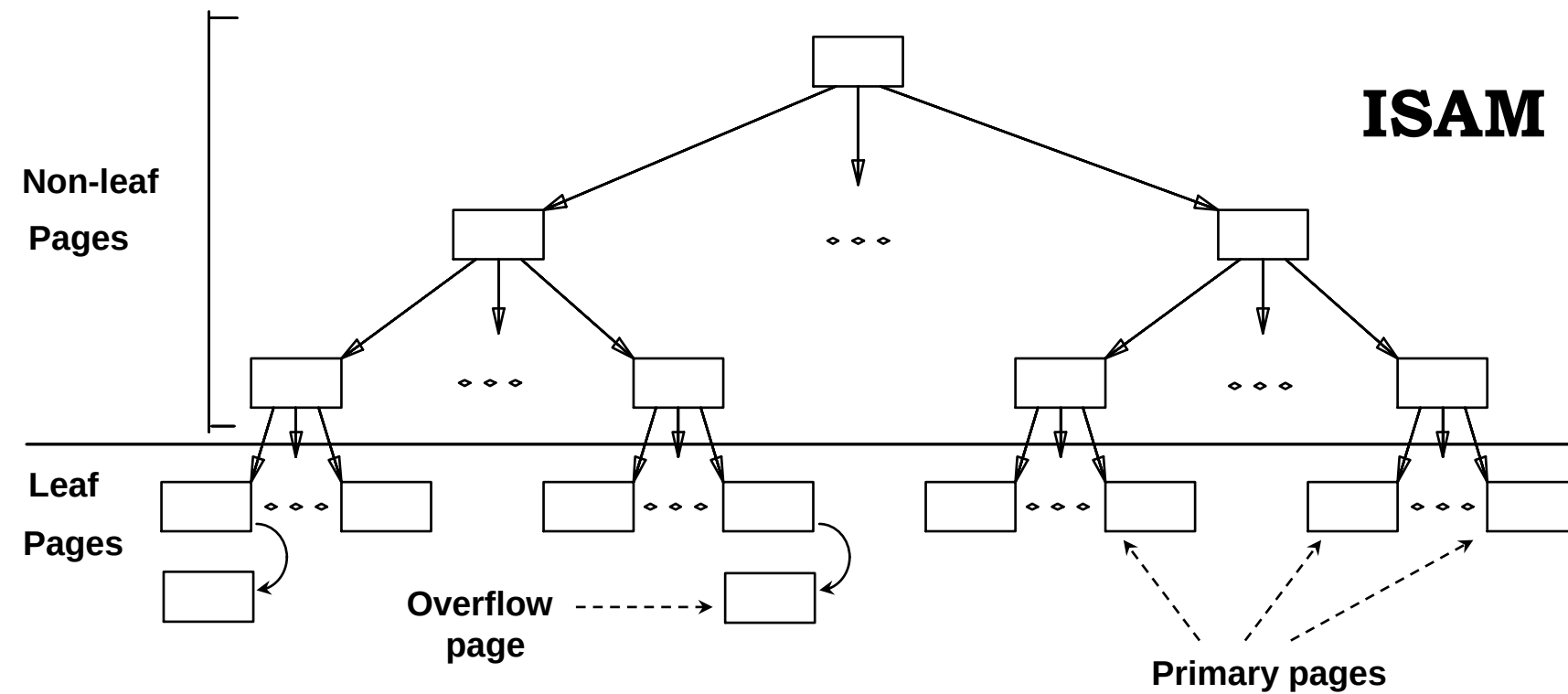
ISAM (INDEX SEQUENTIAL ACCESS METHOD)

Data entries of the ISAM index are in the leaf of the tree and additional overflow pages chained to some leaf pages.

ISAM structure is static (except for overflow pages as they will be very few)



ISAM (INDEX SEQUENTIAL ACCESS METHOD)



ISAM (INDEX SEQUENTIAL ACCESS METHOD)

- Each tree node is disk page.
- When a file is created all leaf pages are allocated sequentially and sorted on the search key value.
- The non leaf level pages are then allocated.
- If there are several inserts to the file (but is there is no space) then additional pages are needed because the index is static (these pages are called Overflow pages).



SUMMARY

- **ISAM**
- **ISAM structure**
- **leaf pages**
- **Non leaf pages**



QUIZ

1-The Oracle RDBMS uses the ____ statement to declare a new transaction start and its properties.

- A) BEGIN
- B) SET TRANSACTION
- C) BEGIN TRANSACTION
- D) COMMIT

Answer: B



2- ____ means that the data used during the execution of a transaction cannot be used by a second transaction until the first one is completed.

- A) Consistency
- B) Atomicity
- C) Durability
- D) Isolation

Answer: D

LECTURE -15



- **ISAM**
- **ISAM structure**
- **leaf pages**
- **Non leaf pages**

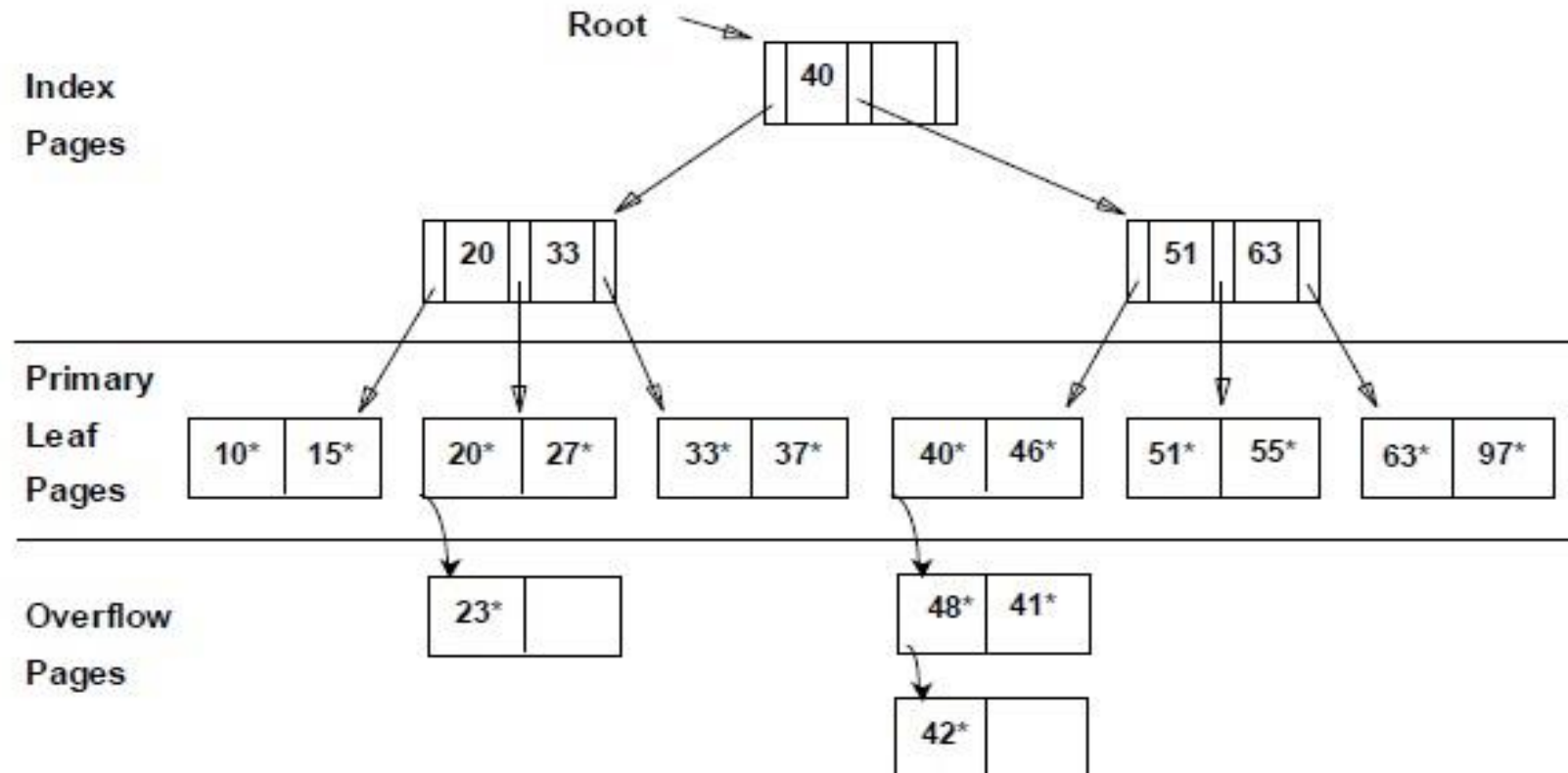
OBJECTIVE

- **B+ Tree**
- **Problem with ISAM**
- **Hash-Based Indexing**



B+ TREES: DYNAMIC INDEX STRUCTURE

- A static structure such as the ISAM index suffers from the problem that long overflow chains can develop as the file grows, leading to poor performance.
- This problem motivated the development of more flexible dynamic structures that



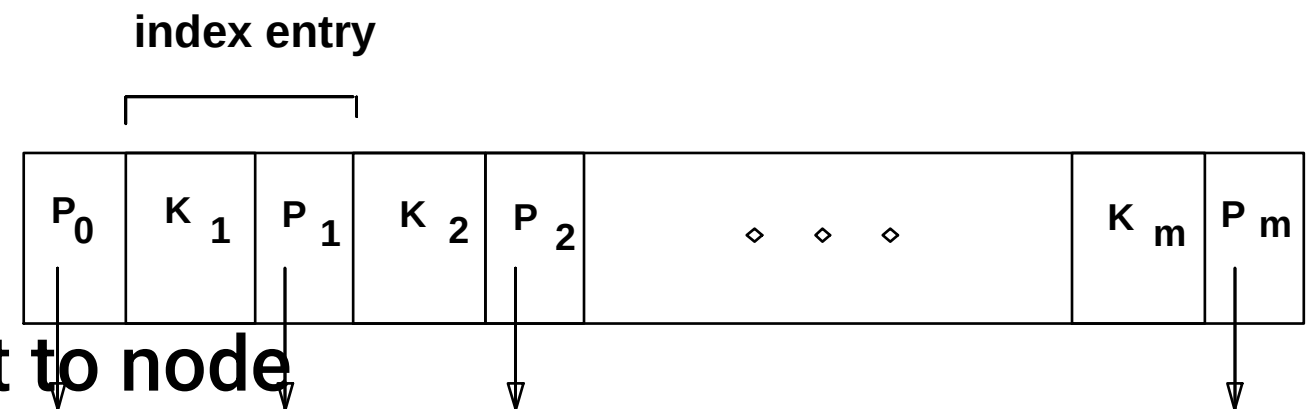
B+ TREES: DYNAMIC INDEX STRUCTURE

- **Problems with ISAM:**

Long overflow pages leads to poor performance

Characteristics:

→ insertion: }
→ Deletion: } In both operations
tree is balanced



Searching: just needs traversal from root to node

Height of the tree: rarely more than 3 to 4

- **Format of the Tree node:** M index entries contains $m+1$ pointers



HASH-BASED INDEXING

- **used to quickly find records that have a given search key value.**
- **Example: if employee records is hashed on the name field, we can retrieve all records about Johnny.**
- **Here records in a file are grouped in buckets**
- **Records from a bucket can be found by using Hash function.**



HASH-BASED INDEXING

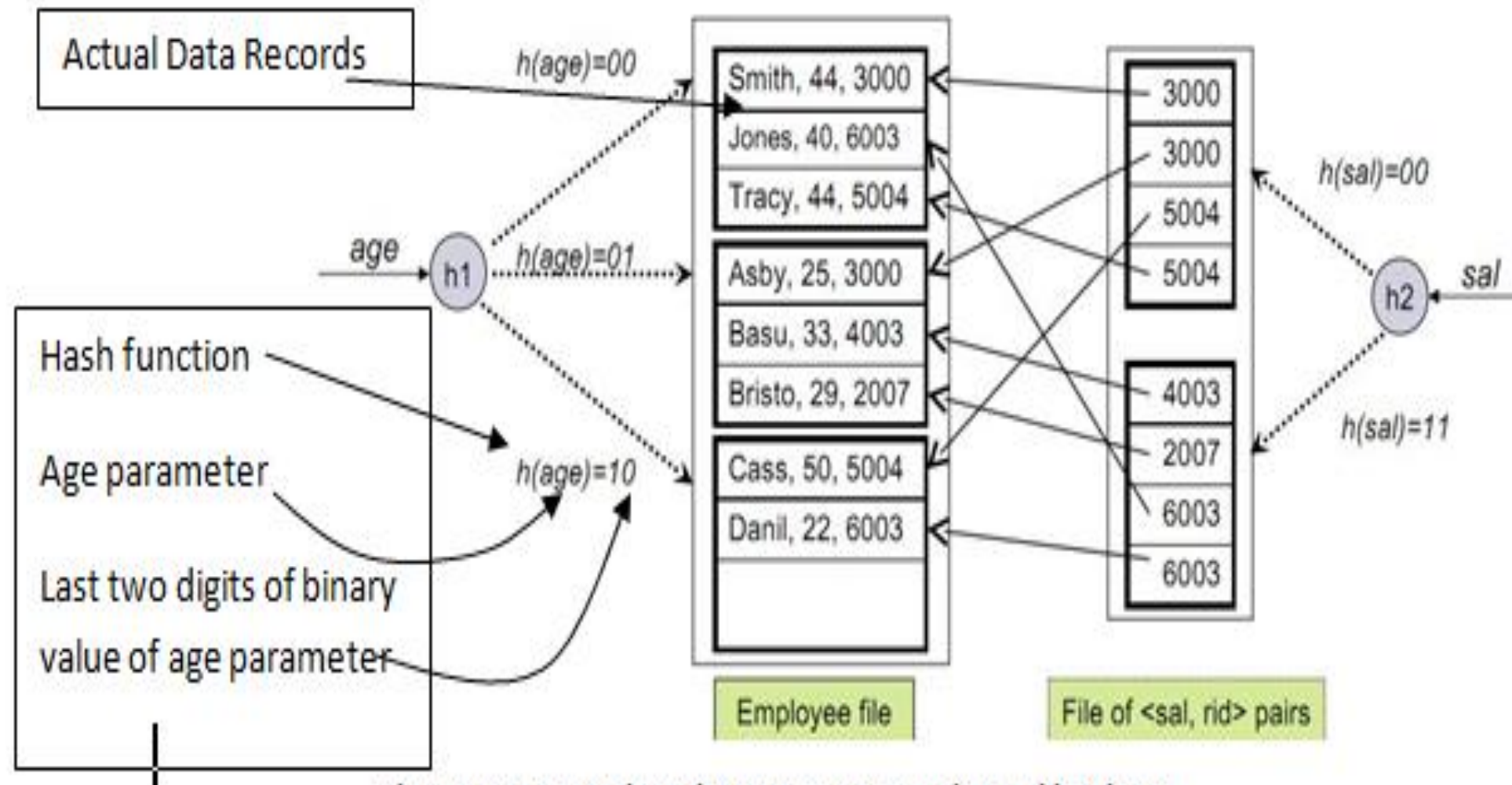


Fig 1.3: Index-organized file hashed on age, with auxiliary index



SUMMARY

- **B+ Tree**
- **Problem with ISAM**
- **Hash-Based Indexing**



QUIZ

1-What are the leaf nodes in a B+ tree? 2-Dynamic hashing is also called as _____

a) The topmost nodes

b) The bottommost nodes

c) The nodes in between the top and bottom nodes

d) None of the mentioned

Answer:b

a) Extended hashing

b) Extendable hashing

c) Static hashing

d) Movable hashing

Answer:b



EXERCISE

A distributed database has which of the following Advantages over a centralized database?

- A) Software cost
- B) Software complexity
- C) Slow Response
- D) Modular growth



Answer: D

EXERCISE

An autonomous homogenous environment is?

- A) The same DBMS is at each node and each DBMS works independently.
- B) The same DBMS is at each node and a central DBMS coordinates database access.
- C) A different DBMS is at each node and each DBMS works independently.
- D) A different DBMS is at each node and a central DBMS coordinates database access.

EXERCISE

A transaction manager is ?

- A) Maintains a log of transactions
- B) Maintains before and after database images
- C) Maintains appropriate concurrency control
- D) All of the above.

Answer: D



EXERCISE

Location transparency allows ?

- A) Users to treat the data as if it is at one location**
- B) Programmers to treat the data as if it is at one location**
- C) Managers to treat the data as if it is at one location**
- D) All of the above.**

Answer: D



EXERCISE

A heterogeneous distributed database is ?

A) The same DBMS is used at each location and data are not distributed across all nodes.

B) The same DBMS is used at each location and data are distributed across all nodes.

C) A different DBMS is used at each location and data are not distributed across all nodes.

D) A different DBMS is used at each location and data are distributed across

THANK YOU